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Department of Electrical and Computer Engineering

ENCS3320 – Computer Networks

Project #2 (Cisco Packet Tracer)

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Abstract

This project involves the design and implementation of a complete IP-based network using Cisco Packet Tracer. The network is divided into five main areas—Core, University, Street, Home, and Datacenter—each with its own subnet and function. Key networking services such as DHCP, DNS, web hosting, email communication, wireless access, and OSPF dynamic routing were configured to simulate a real-world environment.

The network setup included assigning static and dynamic IPs, connecting routers and switches, enabling communication across different areas, and ensuring proper access to web and mail servers. Devices such as PCs, wireless devices, and servers were tested to confirm full connectivity and service availability. Issues such as incorrect gateway assignment and IP conflicts were identified and resolved through troubleshooting.

This project demonstrates the practical application of networking concepts and highlights the importance of planning, and configuration in building a reliable and functional network.

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Theory

Cisco Packet Tracer

Cisco Packet Tracer is a network simulation tool developed by Cisco, designed to facilitate the practice of both simple and complex networking concepts. Cisco advocates learning networking through hands-on experience, and this tool aims to help students grasp networking principles while building skills specific to Cisco technologies.[2]

The primary purpose of Cisco Packet Tracer is to provide students with practical experience in networking. Although it emulates networking protocols using software, it cannot fully replace physical routers or switches. Interestingly, the tool includes not only Cisco devices but also a variety of other networking equipment.

Features of CISCO Packet Tracker

CISCO Packet Tracer supports a multi-user system, enabling users to connect and collaborate on different topologies across various computer networks. This feature allows teachers to assign distinct tasks to students efficiently.

The capabilities of CISCO Packet Tracer can also be modified or restricted using an API, which is another functionality provided by the software. Special features such as accessibility, gaming, assessment delivery, and real-world interaction can be disabled if needed.[2]

The software is available for free download through a Netacad account. Users can simulate router configurations, which are accessible from anywhere and support an unlimited number of devices.

CISCO Packet Tracer offers a self-paced, interactive learning environment. Its Enhanced Physical Mode provides a virtual lab experience for tasks such as cabling devices on a rack, device placement (Rack & Stack), powering devices on and off, managing device port connections, selecting cables, and troubleshooting.

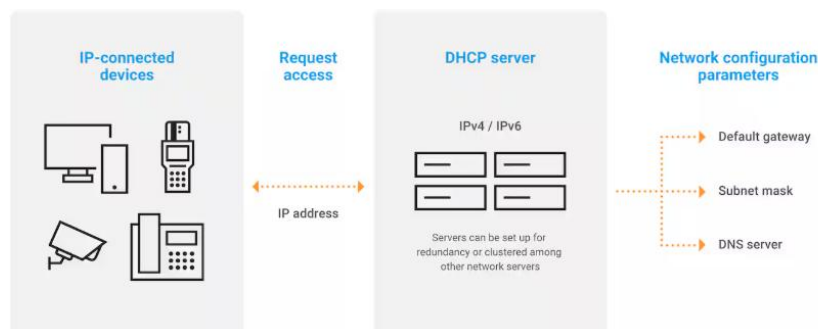
The Network Controller feature offers a centralized dashboard for monitoring the network's status, identifying and resolving issues, and deploying configuration changes to all managed devices simultaneously. This can be done via its Web GUI or

APIs. Additionally, users can utilize real-world applications on their computers to interact with the Network Controller and execute infrastructure automation scripts.

Dynamic Host Configuration Protocol (DHCP)

The Dynamic Host Configuration Protocol (DHCP) is a system used to automatically assign Internet Protocol (IP) addresses to network devices, known as hosts, within an organization's network. These hosts can include desktop computers, laptops, tablets, mobile devices, thin clients, and other types of devices. Each host requires an IP address to communicate with other devices over the internet.[3]

Instead of requiring manual configuration by network administrators, DHCP automatically handles the assignment of IP addresses. It also updates IP addresses when devices change locations within the network. In addition to IP addresses, DHCP provides essential configuration details such as Domain Name System (DNS) addresses, subnet masks, and default gateways to ensure seamless network communication.[3]



How Does DHCP work?

In a TCP/IP-based network, each device requires a unique unicast IP address to access network resources. Without DHCP, IP addresses for new devices or devices moved to a different subnet must be configured manually, and addresses from removed devices must be reclaimed manually as well.[3]

DHCP automates and centralizes this process. A DHCP server manages a pool of IP addresses and assigns an address to any DHCP-enabled client when it connects to the network. These IP addresses are dynamic (leased temporarily) rather than static (permanently assigned), meaning unused addresses are automatically returned to the pool for reuse.[3]

Network administrators configure DHCP servers to maintain TCP/IP settings and provide address assignments to DHCP-enabled clients through lease offers. The DHCP server's database includes:

- **TCP/IP configuration parameters:** Valid settings for all network clients.
- **IP address pool:** A list of valid IP addresses available for assignment, excluding specific reserved addresses.
- **Reserved addresses:** IP addresses consistently assigned to specific DHCP clients for consistent identification.
- **Lease duration:** The period an IP address is valid before requiring renewal.

When a DHCP-enabled client accepts a lease offer, it receives:

- A valid IP address for the subnet it is joining.
- Requested DHCP options, which include additional configuration parameters like the default gateway (router), DNS servers, and DNS domain name.[3]

Importance of DHCP

- **Reliable IP address configuration:** DHCP reduces the risk of configuration errors, such as typographical mistakes or address conflicts that can occur when an IP address is assigned to multiple devices simultaneously.
- **Reduced network administration:** DHCP streamlines network management through several features:
 - Centralized and automated TCP/IP configuration.
 - The ability to define and manage TCP/IP settings from a single location.
 - The capacity to assign a variety of additional TCP/IP configuration values using DHCP options.
 - Efficient handling of IP address updates for devices that frequently change locations, such as mobile devices on a wireless network.
 - The use of a DHCP relay agent to forward initial DHCP messages, removing the need for a separate DHCP server on each subnet.[1]

The components of the Dynamic Host Configuration Protocol

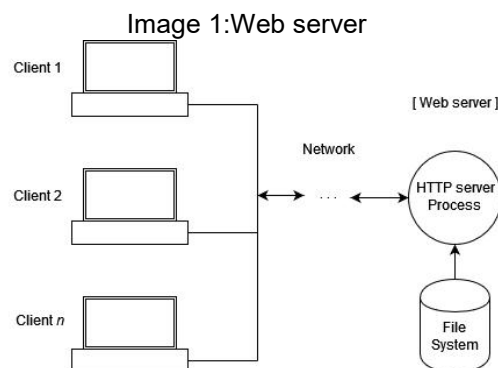
- **Server.**
- **DHCP client.**
- **DHCP relay.**
- **IP address pool.**
- **Subnet.**
- **Lease.**
- **DNS servers.**
- **Default gateway.[4]**

Web Servers

A web server is software and hardware that handles requests via HTTP or HTTPS, responding with requested web pages or resources, or error messages. It can also store resources sent from user agents, such as web browsers or crawlers. The hardware used for web servers varies depending on traffic volume, from embedded systems like routers to large-scale setups with hundreds of high-speed servers for high-traffic websites.[7]

Web server resources can either be static content (pre-existing files) or dynamic content, generated at the time of the request by other programs. Static content is faster to serve and easier to cache, while dynamic content supports more complex applications.[7]

In addition to serving web pages, web servers are now used for computer-to-computer communication through technologies like REST, SOAP, and WebDAV extensions, expanding their role beyond serving human-readable pages.



Importance of Web Servers

A web server processes requests from end users through a web browser, using Hypertext Transfer Protocol (HTTP) for communication. The web server's main function is to store, process, and deliver requested web pages or information. It works as follows:

- **Physical Storage:** Website data is stored on a physical server for security. When a user enters a URL or searches for a site, a request is sent to the web server to retrieve the data.[7]
- **Web Browser:** Browsers like Firefox, Chrome, or Internet Explorer locate the web server where the website data is stored. After finding the server, the browser reads and processes the request to display the information.

What are web servers used for?

Web servers are primarily responsible for handling and managing HTTP/HTTPS requests and responses from client systems.

In addition to this core function, a web server can perform several other tasks, including:

- Storing and protecting website data: Web servers can safeguard critical website data from unauthorized access.
- Managing bandwidth to control network traffic: Web servers help prevent downtime due to high web traffic by allowing web hosts to regulate data transmission rates and reduce excessive network congestion.[7]
- Server-side web scripting: This feature enables the creation of dynamic web pages using scripting languages such as Ruby, Python, and PHP.
- Virtual hosting: Web servers can also act as virtual servers, hosting multiple applications, websites, data, and other services.

DNS

The Domain Name System (DNS) is an important component of the internet, as it translates human readable domain names into machine readable IP addresses. This translation enables users to access websites using easy to remember domain names instead of numerical IP addresses .

How DNS Works:

1. **DNS Query Initiation:** When a user enters a domain name (e.g., www.google.com) into a web browser, the browser initiates a DNS query to resolve this domain name into an IP address .
2. **Proxy server:** The query is first sent to a proxy server, typically managed by the user's Internet Service Provider (ISP). If the resolver has the IP address cached, it returns the result; otherwise, it queries other DNS servers.
3. **Root Name Servers:** The resolver contacts a root name server, which directs it to the appropriate Top-Level Domain (TLD) name server based on the domain's extension [4].
4. **TLD Name Servers:** The resolver queries the TLD name server, which responds with the authoritative name server for the specific domain [4].
5. **Authoritative Name Server:** The authoritative name server provides the corresponding IP address for the requested domain name [4].
6. **Response to Client:** The proxy server returns the IP address to the user's browser, which can then establish a connection to the web server [4].

DNS Components:

- **Domain Names:** Structured hierarchically, domain names consist of multiple labels separated by dots, read from right (top-level domain) to left (subdomains) [4].
- **Name Servers:** Servers that store DNS records and are responsible for answering queries about domain names, including root name servers, TLD name servers, and authoritative name servers [4].
- **DNS Records:** Entries that provide information about a domain, we learned some of them in our course such as A, CNAME, NS records [5].

DNS Record Types

1. A Record (Address Record):

The A record maps a domain name to an IPv4 address, which is a numerical address used by computers to identify servers on a network. For example, if you enter example.com into your browser, the A record might resolve it to an

IP address like 192.168.1.1. This record is fundamental for connecting domain names to web servers.

- Example: example.com → 192.168.1.1 [5]

2. CNAME Record (Canonical Name Record):

The CNAME record creates an alias for a domain name, pointing it to another domain name rather than an IP address. This is useful when multiple domain names need to point to the same resource. For example, www.example.com can be set as a CNAME pointing to example.com, so both resolve to the same website without needing separate A records.

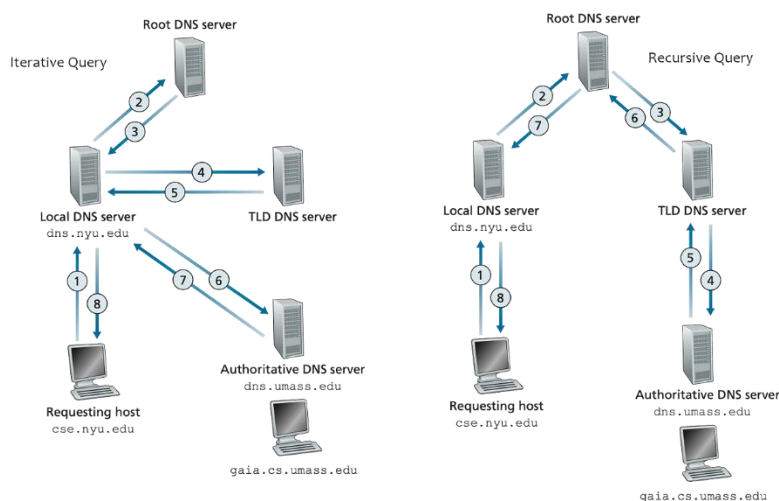
- Example: www.example.com → example.com [4]

3. NS Record (Name Server Record):

The **NS record** specifies the authoritative name servers for a domain. These are the servers responsible for responding to DNS queries about that domain. For instance, if you register example.com, your registrar will configure NS records pointing to your hosting provider's name servers. This ensures the correct name servers handle all DNS lookups for the domain.

- Example: example.com → ns1.examplehost.com [4]

DNS inquiry Types:



- **Recursive Resolution:** The DNS resolver queries multiple DNS servers to resolve the domain name. [5]

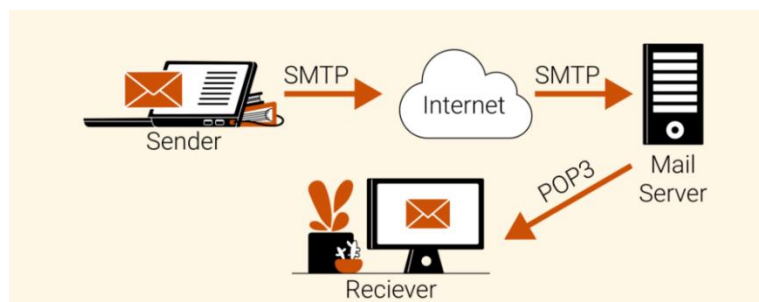
- **Iterative Resolution:** The resolver queries each server sequentially, with each server referring it to the next closer to the desired IP address [8].

Security Considerations:

DNS was not originally designed with security in mind, making it open to attacks such as DNS spoofing or cache poisoning. DNS Security Extensions (DNSSEC) ensure data integrity and authenticity by allowing DNS responses to be verified [9].

Email Servers

Email communication relies on of protocols that secure the sending, receiving, and storage of messages. The main protocols involved are SMTP for sending emails, and POP3 and IMAP for retrieving them as we learned in the ENCS3320 course.



Simple Transfer Protocol (SMTP):

Mail

SMTP is the standard protocol for sending emails across networks. It operates over TCP port 25 for communication between mail servers and port 587 for mail submission from clients. As seen in the figure above when a user sends an email, the email client (Mail User Agent or MUA) forwards the message to a Mail Submission Agent (MSA) using SMTP. The MSA then sends the email to the recipient's mail server (Mail Transfer Agent or MTA), which delivers it to the recipient's mailbox. SMTP is designed mainly for sending messages and is not for retrieving or storing emails [5].

Post Office Protocol version 3 (POP3):

POP3 is a protocol used by email clients to retrieve messages from a mail server. Operating over TCP port 110, POP3 downloads emails from the server to the user's device and typically deletes them from the server afterward. This allows users to access their emails offline but limits access to the same messages from multiple devices, as the emails stay only on the device to which they were downloaded [5].

Interaction Between SMTP and POP3:

In a typical email, SMTP handles the transmission of emails from the sender to the recipient's mail server, while POP3 or IMAP manages the retrieval of emails from the server to the recipient's device. This division of labor ensures efficient delivery and access to email messages [5].

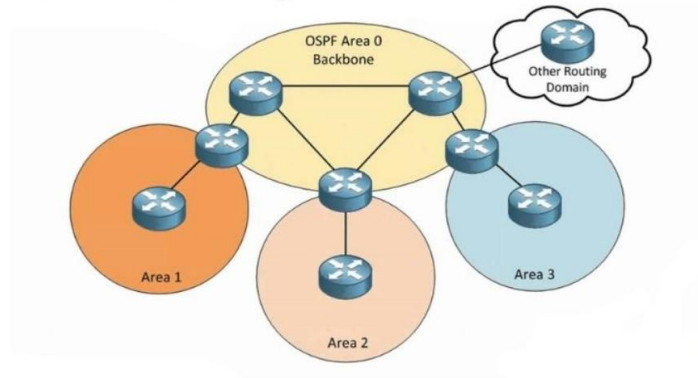
Security Considerations:

Both SMTP and POP3 can operate over secure connections using TLS (Transport Layer Security) to encrypt the communication between clients and servers, protecting sensitive information from potential eavesdropping. It's recommended to use the secure ports (587 for SMTP submission, 995 for POP3, and 993 for IMAP) to ensure data privacy .

Open Shortest Path First (OSPF)

Open Shortest Path First (OSPF) is an interior gateway protocol (IGP) for routing within an autonomous system (AS). It operates using a link state routing algorithm, enabling routers to dynamically exchange information about the network's topology and compute the shortest paths to destinations. OSPF is designed for scalability, fast convergence, and efficient routing, making it suitable for complex networks [6].

OSPF Hierarchy



Key Features of OSPF

OSPF uses Link-State Advertisements (LSAs) to share information about local connections and network state. These LSAs are propagated to all routers within the same area, ensuring a consistent view of the network topology. To manage large networks, OSPF supports a hierarchical structure by dividing the network into areas. A backbone area (Area 0) serves as the core, connecting all other areas, which reduces routing overhead and improves scalability [6].

Each OSPF interface is assigned a cost metric, usually based on bandwidth. Routers compute the shortest path to a destination by summing the costs of the links along a route, selecting the path with the lowest total cost. OSPF routers establish neighbor relationships through Hello packets, forming adjacencies with selected neighbors. The neighbor states include Down, Init, 2-Way, Exstart, Exchange, Loading, and Full. In the Full state, routers are fully adjacent, and their link state databases are synchronized [14].

In multi-access networks, such as Ethernet, OSPF elects a Designated Router (DR) and a Backup Designated Router (BDR) to streamline LSA dissemination and reduce the number of adjacencies. This election improves efficiency and reduces network traffic [6].

OSPF Packets and Convergence

OSPF uses five packet types to perform its functions:

- **Hello Packets:** to discover and maintain neighbor relationships.

- Database Description (DBD) Packets: to summarize a router's database during the exchange process.
- Link-State Request (LSR) Packets: to request specific LSAs.
- Link-State Update (LSU) Packets: to send LSAs to neighbors.
- Link State Acknowledgment (LSAck) Packets: to confirm receipt of LSAs .
-

OSPF is known for its fast convergence. When a network change occurs, routers propagate updated LSAs, and the Shortest Path First (SPF) algorithm recalculates the optimal paths. This rapid response minimizes disruptions in the network. Additionally, OSPF supports authentication mechanisms to secure routing exchanges, preventing unauthorized devices from getting routing data [6].

Procedure

3.1 Internet Protocol (IP) Addressing and Subnetting

2.1.1 Subnetting Concepts

Network devices use IP addresses and subnets to identify the source and destination of communications and manage network addresses respectively. IP addresses contain two parts: a network identifier and a host identifier. The network ID specifies an area of the network where a device resides, much like an area code identifies a section of a telephone network. The host ID labels a specific device in that network section, similar to how a telephone number identifies a specific phone within an area code.

Most business networks still rely on IP version 4 (IPv4) addresses, which offer about 4.3 billion unique variations. Internet identities use most of these, but the newer IP version 6 (IPv6) standard provides more addresses and other benefits.

Computers work with IPv4 addresses as 32-bit binary strings. Humans usually convert binary to a dotted decimal address, which is easier to write and understand. For example, the binary string 11000000.00000000.00000010.00000010 converts to the IP address 192.0.2.2. Its associated subnet mask converts from 11111111.11111111.11111111.11111100 to 255.255.255.252.

Subnet masks clarify which part of the address is the network ID and which is the host ID. Routers, computers and network troubleshooters use IP addresses and subnet masks to manage network traffic, which ensures information sent from one system arrives at its destination.[1]

2.1.2 Procedure

A main IP address block of: 1X.Y.8.0/23

was assigned for the entire network. The values of **X** and **Y** were determined using the last four digits of the team leader's student ID , In this project, the team leader's SID is **1221054**, so:

$$X = 10$$

$$Y = 54$$

As a result, the base network used was:

$$110.54.8.0/23$$

This address block provides a total of **510 usable IP addresses**, which were divided into multiple subnets according to the network topology requirements. The topology consists of five areas, all of which are connected through the core network. Each area was allocated a subnet based on the number of hosts needed.

The following table shows the allocation:

Area	Subnetwork	# Hosts Needed	Subnet Mask (CIDR)	Subnet Mask (Decimal)	Network Address	First Usable IP	Last Usable IP	Broadcast Address
University	NET1-A	60	/26	255.255.255.192	110.54.8.0	110.54.8.1	110.54.8.62	110.54.8.63
	NET1-B	60	/26	255.255.255.192	110.54.8.64	110.54.8.65	110.54.8.126	110.54.8.127
Street	NET2	30	/27	255.255.255.224	110.54.8.128	110.54.8.129	110.54.8.158	110.54.8.159
Home	NET3	20	/27	255.255.255.224	110.54.8.160	110.54.8.161	110.54.8.190	110.54.8.191
Datacenter	NET4	15	/27	255.255.255.224	110.54.8.192	110.54.8.193	110.54.8.222	110.54.8.223
Core	NET0-A	2	/30	255.255.255.252	110.54.8.224	110.54.8.225	110.54.8.226	110.54.8.227
	NET0-B	2	/30	255.255.255.252	110.54.8.228	110.54.8.229	110.54.8.230	110.54.8.231
	NET0-C	2	/30	255.255.255.252	110.54.8.232	110.54.8.233	110.54.8.234	110.54.8.235

Table 1:subnetting table

2.1.3. Address Assignment Strategy

To efficiently assign IP addresses, subnets were arranged in descending order of host requirements. This ensured that larger subnets were allocated first, avoiding overlaps and preserving address space.

The University Network required the most hosts, so two /26 subnets were assigned first. Next, the Street, Home, and Datacenter networks were each given a /27 subnet. Finally, three /30 subnets were reserved for the Core Network to connect routers.

This approach provided the required number of IPs for each area while optimizing usage within the 110.54.8.0/23 block.

Different **CIDR subnet masks** were selected based on the number of hosts required in each network:

- A **/26** subnet (62 usable IPs) was used for NET1-A and NET1-B in the University Network, as each needed around **60 hosts**.
- A **/27** subnet (30 usable IPs) was assigned to the Street, Home, and Datacenter Networks, where host requirements ranged from **15 to 30**.
- A **/30** subnet (2 usable IPs) was chosen for Core Network links (NET0-A to NET0-C), as each point-to-point router connection only required **2 addresses**.

By matching the subnet size to the exact number of needed hosts, address space was used efficiently and without waste.

Network Topology Design

Overview of Network Structure

The network topology was designed to represent a realistic and scalable communication system. It was divided into five main areas, each serving a specific function:

Core Network (Area 0):

Acts as the backbone of the topology. All other areas are interconnected through this central hub. It enables routing between different parts of the network using dynamic protocols.

University Network (Area 1):

Contains PCs, servers, and wireless devices simulating a campus environment. Two separate subnets were created within this area to handle both wired and wireless traffic efficiently.

Street Network (Area 2):

Includes a Cell Tower, Central Office Server, and smart devices like smartphones, reflecting a public urban access scenario. Wireless connectivity was configured to simulate real-world street-level internet access.

Home Network (Area 3):

Simulates a small home environment with end-user PCs and a local router. This area is used to test access to services hosted in other network segments like the datacenter.

Datacenter Network (Area 4):

Hosts essential services including a Web Server, DNS Server, and Email Server. These are accessed by devices from all other areas to test internal and external communication.

Each of these networks was connected to the Core Router using point-to-point links. The entire topology was implemented using Cisco Packet Tracer, following logical IP planning and structured subnetting.

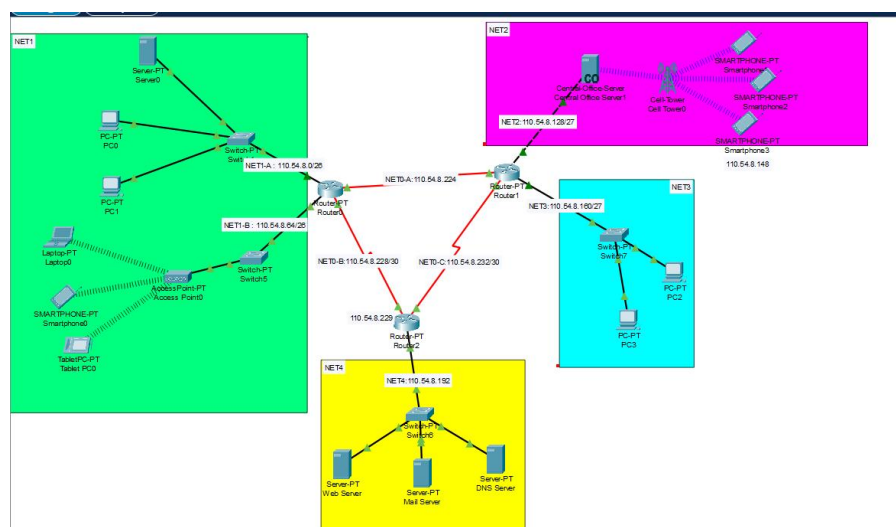


Figure 1: main topology

Core Network (Area 0):

The Core Network, also referred to as Area 0, serves as the backbone of the entire network topology. It is responsible for interconnecting all other network areas, including the University, Street, Home, and Datacenter networks. All routing traffic between these areas is managed and forwarded through the Core.

To ensure efficient and structured routing, OSPF (Open Shortest Path First) was configured using a hierarchical multi-area design, with Area 0 acting as the backbone area. All other areas are connected to this core, enabling optimized path calculation and reduced routing overhead.

In Area 0, three point-to-point subnetworks were assigned:

NET0-A: Connects Router0 and Router1.

NET0-B: Connects Router1 and Router2.

NET0-C: Connects Router1 and Router3.

Each of these subnets uses a /30 CIDR block (255.255.255.252), which provides exactly two usable IP addresses — ideal for point-to-point links between routers, as no additional hosts are needed.

By isolating inter-area routing in the Core, a clear separation of concerns was achieved, which enhanced both network scalability and fault isolation.

Router0

Router0 Connects the University Network (Area 1) to the Core

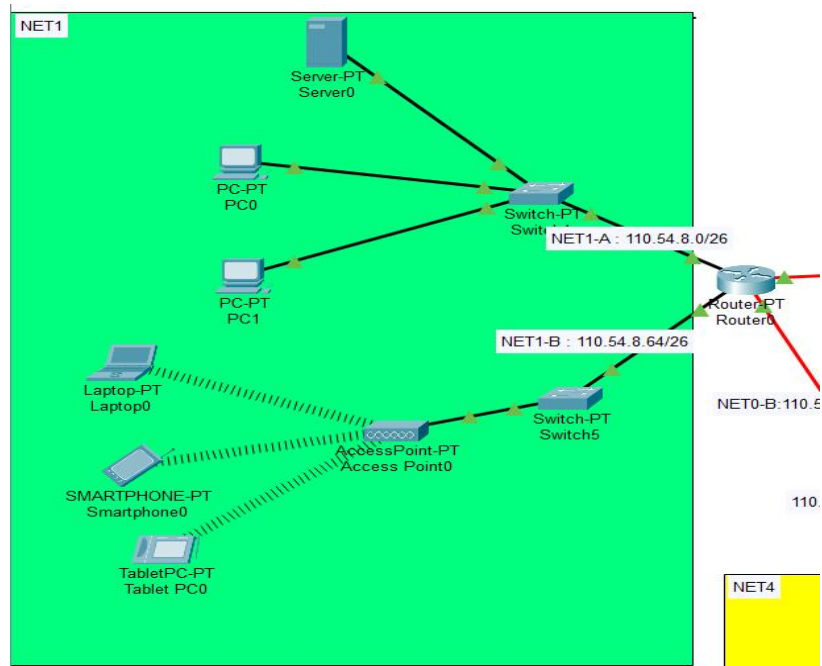


Image 2:Router0-connections

Router0 – Interface Configuration Summary

Interface	IP Address	Subnet Mask	Connected To	Subnet	Area
GigabitEthernet6/0	110.54.8.1	255.255.255.192	Switch1 (wired hosts in NET1-A)	NET1-A	Area 1
Serial2/0	110.54.8.225	255.255.255.252	Router1 (Core backbone)	NET0-A	Area 0
Serial3/0	110.54.8.230	255.255.255.252	Router2 (Core backbone)	NET0-B	Area0
GigabitEthernet7/0	110.54.8.65	255.255.255.192	Wireless Router (for NET1-B)	NET1-B	Area 1

Table 2:Router0-interfaces

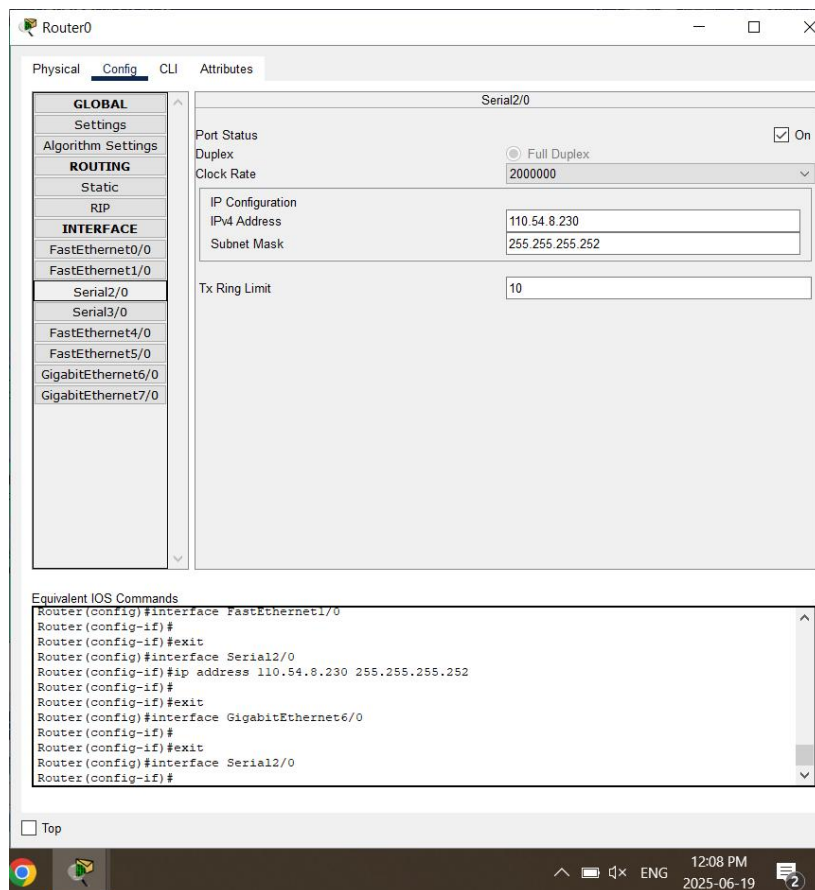


Image 3: Router0-one interface config example

Each interface was configured with the appropriate static IP and subnet mask. OSPF was enabled on SE2/0 and SE3/0 for Area 0, and on GE6/0 and GE7/0 for Area 1.

OSPF Configuration on Router0

On Router0, Open Shortest Path First (OSPF) was configured to enable dynamic routing between the University Network (Area 1) and the Core Network (Area 0). This configuration allows Router0 to exchange routing information with other routers and ensure full reachability across the topology.

Areas Used:

- **Area 0 (Backbone):** Handles the point-to-point connection to Router1 via Serial3/0 and Serial2/0.

- **Area 1** (University): Advertises internal University subnets connected through GigabitEthernet6/0 and GigabitEthernet7/0.

OSPF Configuration Commands

The following commands were used to configure OSPF on Router0:

```
Router(config)# router ospf 1
```

```
Router(config-router)# router-id 110.54.8.230
```

```
Router(config-router)# network 110.54.8.0 0.0.0.63 area 1
```

```
Router(config-router)# network 110.54.8.64 0.0.0.63 area 1
```

```
Router(config-router)# network 110.54.8.192 0.0.0.3 area 0
```

```
Router(config-router)# network 110.54.8.224 0.0.0.3 area 0
```

The **router-id** was set to uniquely identify Router0 in the OSPF process. Wildcard masks were used to match the exact subnets on each interface. OSPF automatically recognized the active interfaces and began exchanging routing information with Router1 and Router2 and the rest of the OSPF domain.

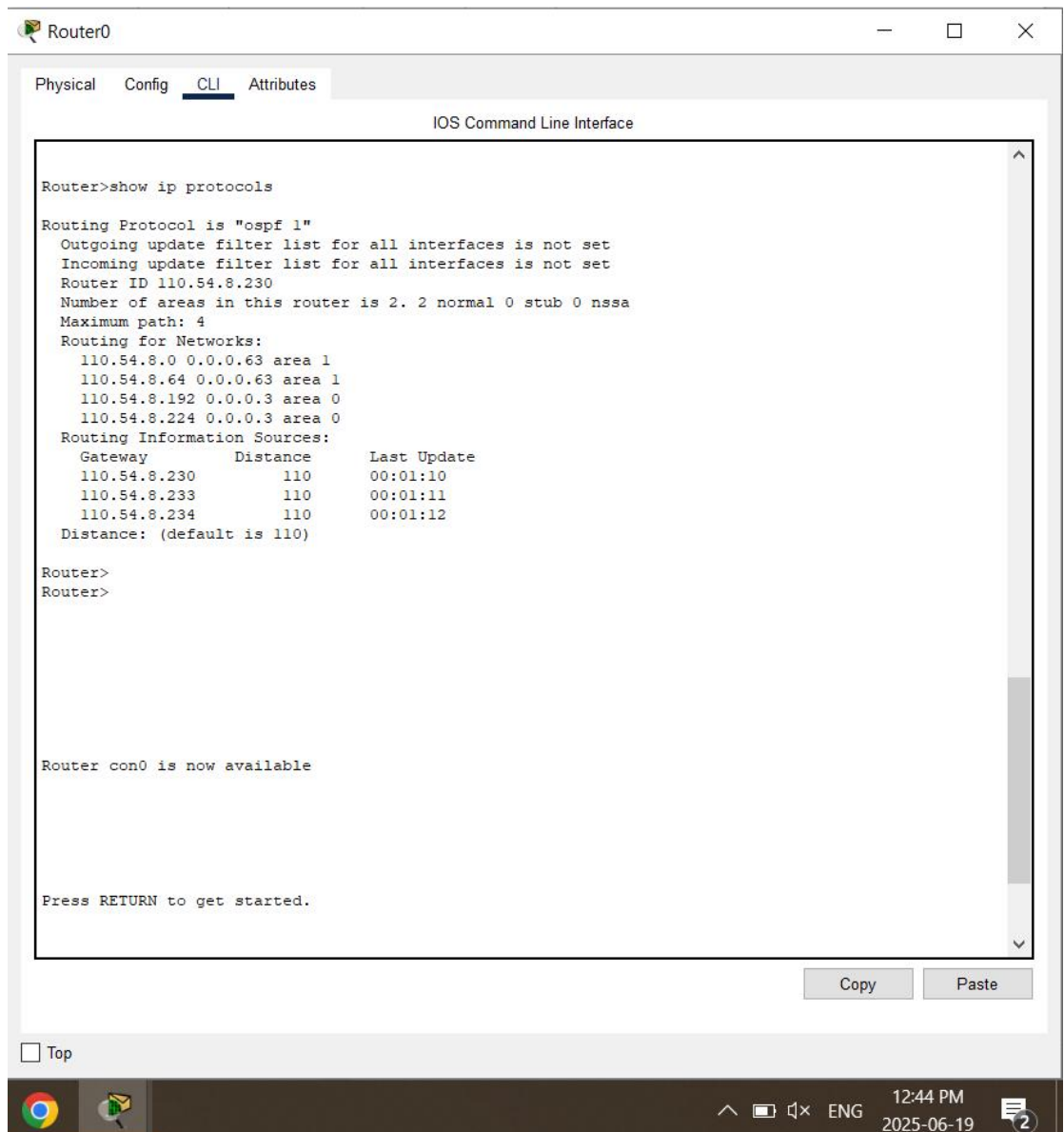


Image 4:Router0-OSPF Configuration

This screenshot displays the OSPF setup on Router0, captured using the show ip protocols command. The router ID is 110.54.8.230, and OSPF is running in two areas:

- **Area 1**, for the University subnets (NET1-A and NET1-B)
- **Area 0**, for the core network links to Router1 and Router2

The bottom part confirms that Router0 is exchanging routing updates with the other routers in the OSPF domain, proving that dynamic routing is working as expected.

Router1

Connects the Street Network (Area 2) and the Home Network (Area 3) to the Core Network (Area 0).

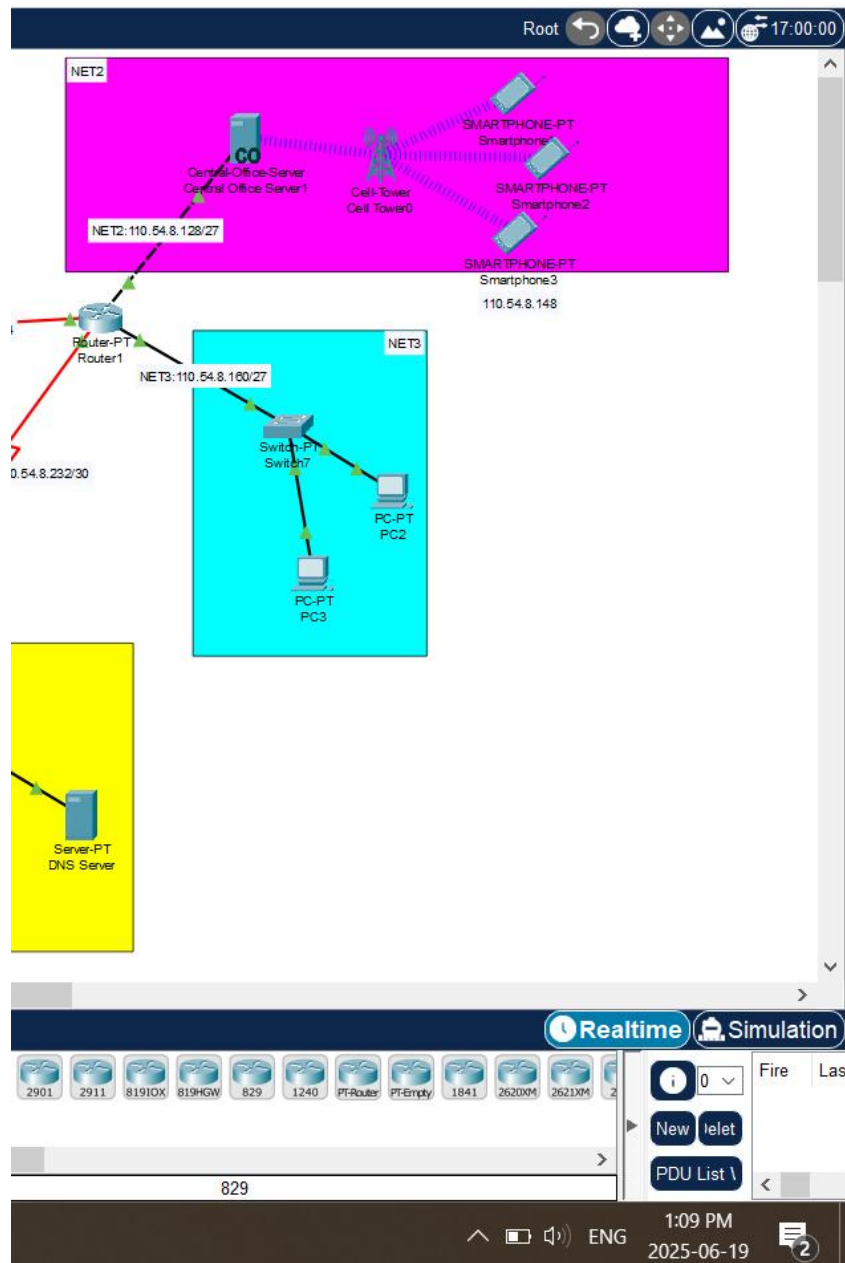


Image 5:Router1-connections

Router1 – Interface Configuration Summary

Interface	IP Address	Subnet Mask	Connected To	Subnet	Area
FastEthernet0/0	110.54.8.129	255.255.255.224	Central Office Server / Cell Tower (NET2)	NET2	Area 2
FastEthernet1/0	110.54.8.161	255.255.255.224	Switch7 (PC2, PC3 in NET3)	NET3	Area 3
Serial2/0	110.54.8.226	255.255.255.252	Router0 (Core)	NET0-A	Area 0
Serial3/0	110.54.8.233	255.255.255.252	Router2 (Core)	NET0-C	Area 0

Image 6:Router1-configurations table

Each interface was configured with a static IP and correct subnet mask. NET2 and NET3 are both served by Router1 as edge networks, while Serial links connect it to the Core.

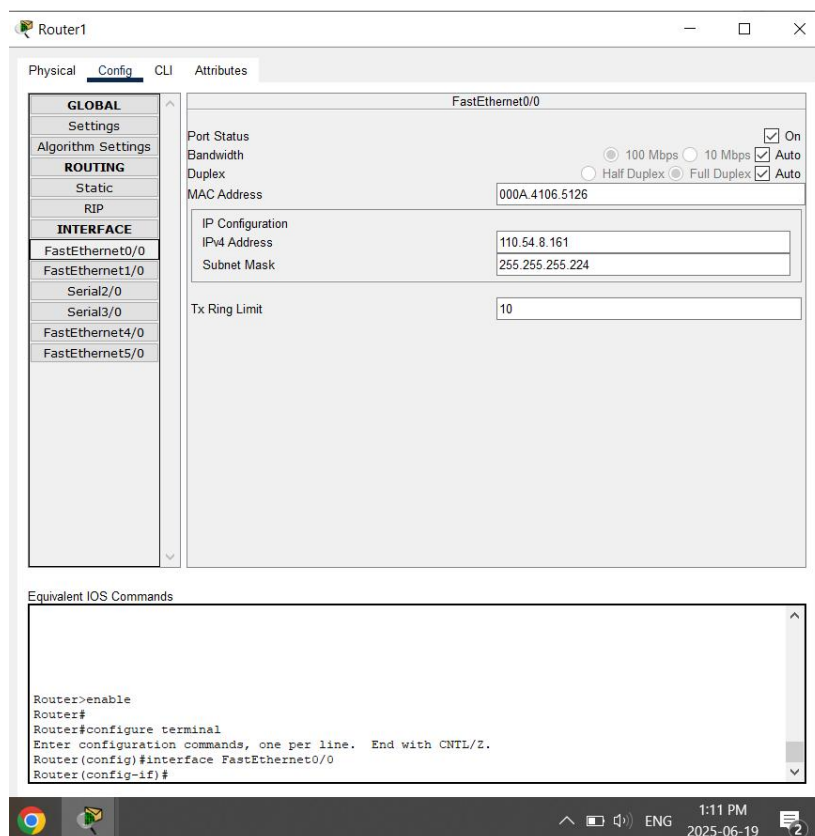


Image 7:Router1-interface configuration example

OSPF Configuration on Router1

OSPF was enabled to allow dynamic routing across the network and ensure Router1 communicates efficiently with both core and edge devices.

Areas Used:

- **Area 0 (Core):** Links to Router0 and Router2 via Serial2/0 and Serial3/0
- **Area 2 (Street):** Central Office Server and Cell Tower
- **Area 3 (Home):** PCs connected via Switch7

OSPF Configuration Commands:

```
Router(config)# router ospf 1
Router(config-router)# router-id 110.54.8.233
Router(config-router)# network 110.54.8.128 0.0.0.31 area 2
Router(config-router)# network 110.54.8.160 0.0.0.31 area 3
Router(config-router)# network 110.54.8.224 0.0.0.3 area 0
Router(config-router)# network 110.54.8.232 0.0.0.3 area 0
```

The router-id uniquely identifies Router1. OSPF began advertising all connected subnets, allowing seamless communication with other routers.

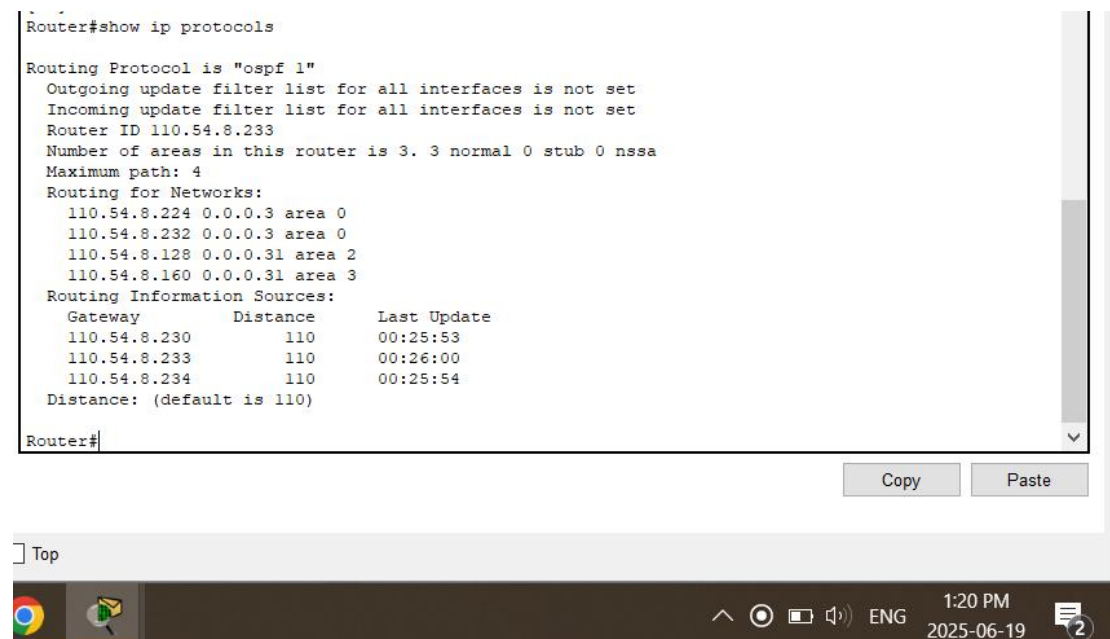


Image 8:Router1-OSPF Configuration

This figure shows the result of the show ip protocols command on Router2. It confirms that OSPF is running with Router ID 110.54.8.233, and that three OSPF areas are being used: Area 0 (core), Area 2 (Street Network), and Area 3 (Home Network). The listed networks match the connected interfaces, meaning OSPF is set up correctly. The lower part shows that Router2 is successfully connected to Router0 and Router1 using OSPF, and they are sharing routing information.

Router2

Router2 connects the Datacenter Network (Area 4) to the Core Network (Area 0). It is linked to Router0 and Router1 via point-to-point serial interfaces and connects internally to the datacenter's servers through a GigabitEthernet interface.

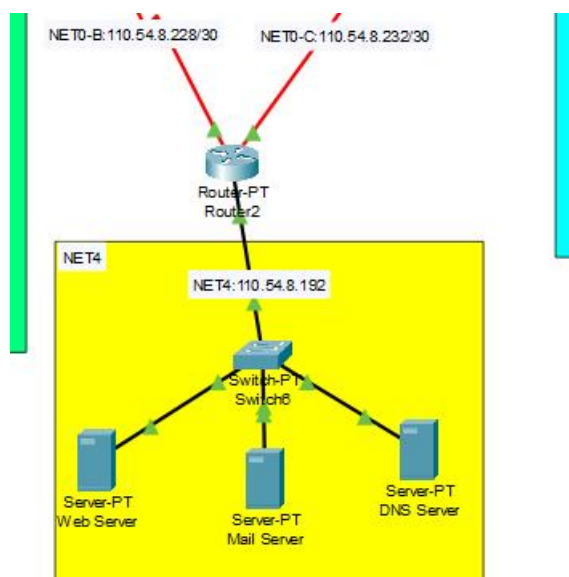


Image 9:Router2-connections

Router2 – Interface Configuration Summary

Interface	IP Address	Subnet Mask	Connected To	Subnet	Area
Serial2/0	110.54.8.229	255.255.255.252	Router0 (NET0-B)	NET0-B	Area 0
Serial3/0	110.54.8.234	255.255.255.252	Router1 (NET0-C)	NET0-C	Area 0

Interface	IP Address	Subnet Mask	Connected To	Subnet	Area
GigabitEthernet6/0	110.54.8.193	255.255.255.224	Switch8 (Datacenter)	NET4	Area 4

Image 10:Router2 – Interface Configuration

Each interface was assigned a static IP address and subnet mask. Serial links are used for inter-router connections in the core, while the GigabitEthernet interface connects to internal services.

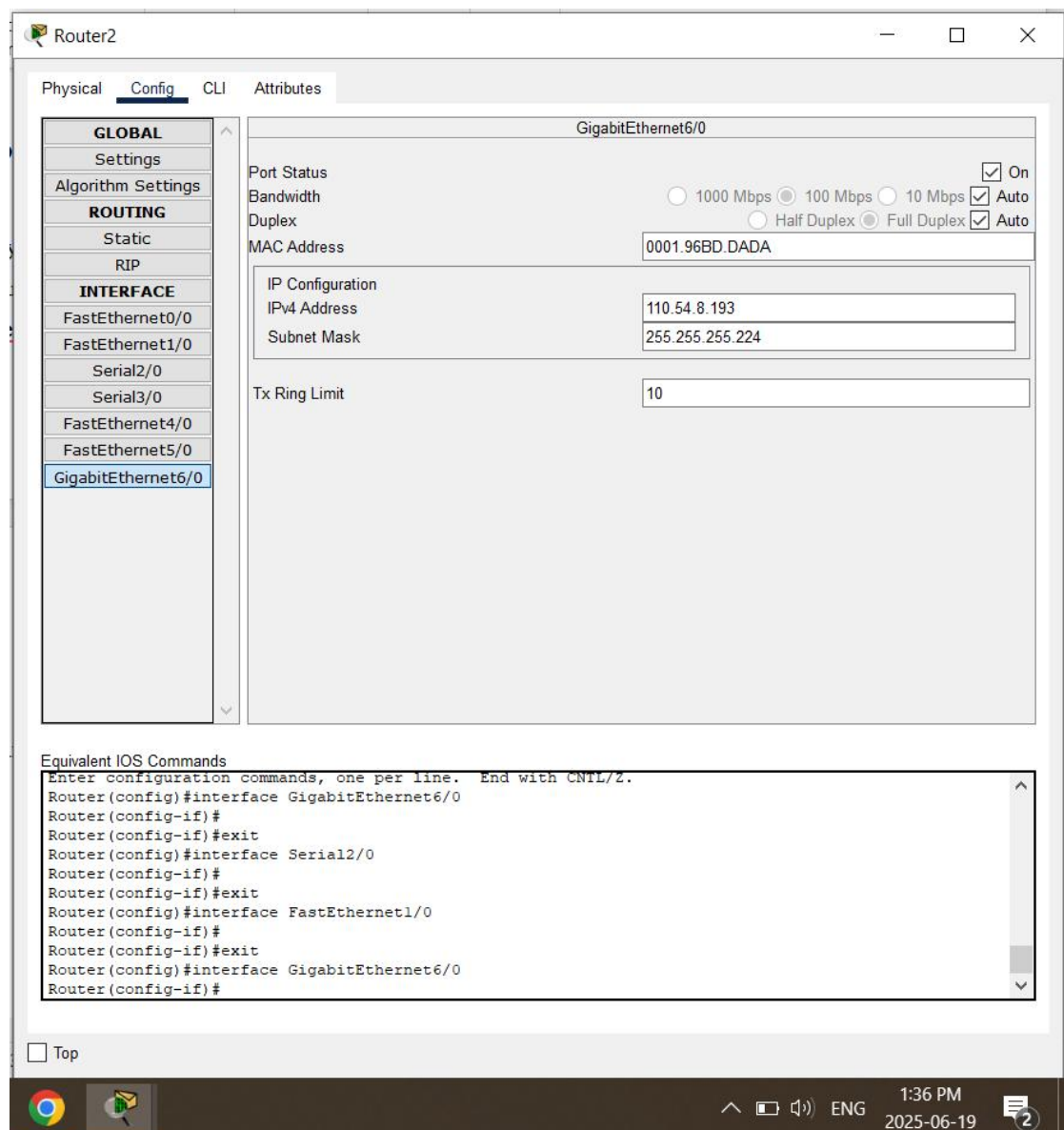


Image 11:Router2-one interface config

OSPF Configuration on Router2

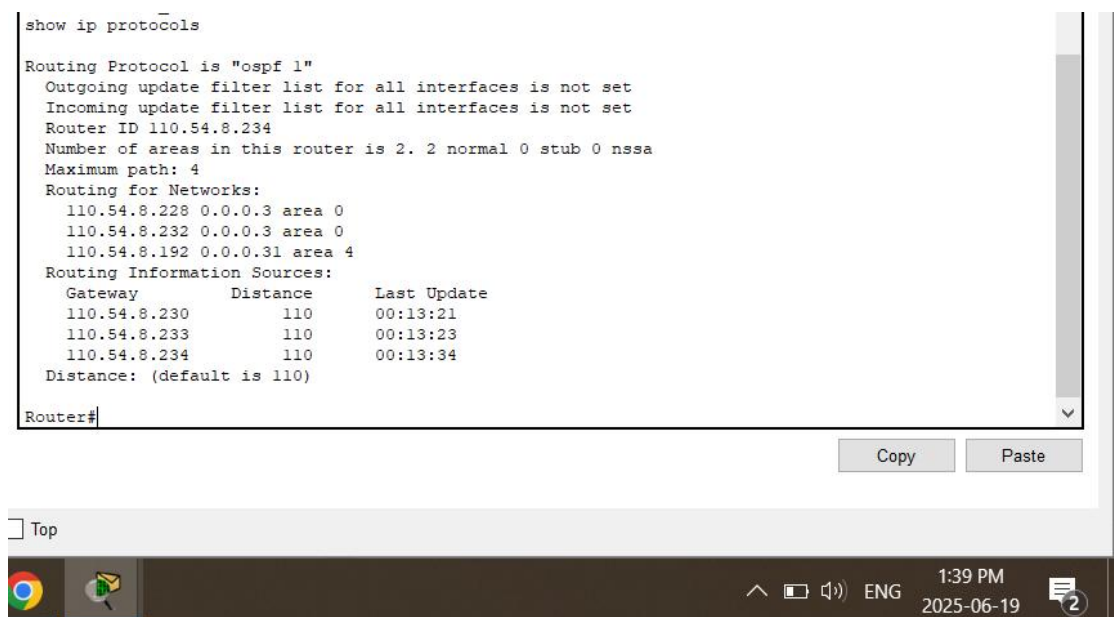
Open Shortest Path First (OSPF) was configured on Router2 to support dynamic routing between the Datacenter Network and the rest of the topology through Area 0 (Core) and Area 4 (Datacenter). This enables full route advertisement and reachability.

Areas Used:

- **Area 0:** Handles communication with Router0 and Router1 via Serial2/0 and Serial3/0
- **Area 4:** Advertises the datacenter servers (Web, Mail, and DNS) through GigabitEthernet6/0

OSPF Configuration Commands:

```
Router(config)# router ospf 1
Router(config-router)# router-id 110.54.8.234
Router(config-router)# network 110.54.8.229 0.0.0.3 area 0
Router(config-router)# network 110.54.8.232 0.0.0.3 area 0
Router(config-router)# network 110.54.8.192 0.0.0.31 area 4
```



```
show ip protocols
Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 110.54.8.234
  Number of areas in this router is 2. 2 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    110.54.8.228 0.0.0.3 area 0
    110.54.8.232 0.0.0.3 area 0
    110.54.8.192 0.0.0.31 area 4
  Routing Information Sources:
    Gateway         Distance      Last Update
    110.54.8.230          110          00:13:21
    110.54.8.233          110          00:13:23
    110.54.8.234          110          00:13:34
  Distance: (default is 110)
Router#
```

Image 12:OSPF Configuration on Router2

Router2 runs OSPF with Router ID 110.54.8.234 and advertises networks in Area 0 (core links to Router0 and Router1) and Area 4 (Datacenter). The show ip protocols command confirms OSPF is active, and routing updates are successfully exchanged with neighboring routers.

University Network (Area 1)

The University Network is divided into two subnets: NET1-A for wired connections and NET1-B for wireless devices. These subnets are routed through Router0, and all IP addressing was configured to support connectivity and integration with OSPF Area .

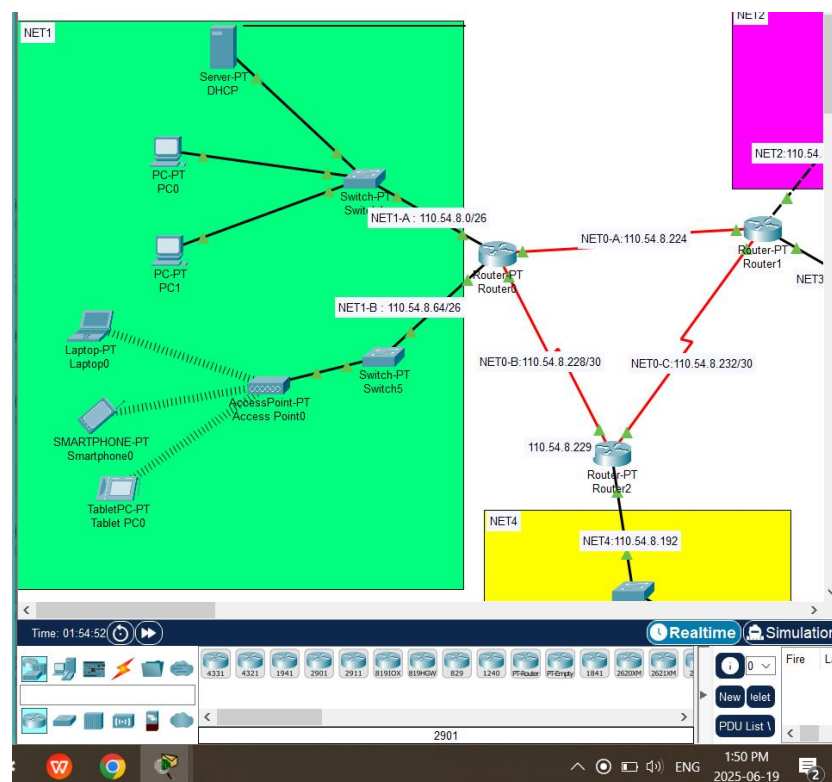


Image 13:University Network connections

1. NET1-A – Wired Network

Components:

- **Switch1** – Distributes connectivity to end devices.

- **PC0 & PC1** – Wired desktop clients.
- **DHCP Server** – Provides dynamic IP addressing for wired hosts.
- **Router0 (GigabitEthernet6/0)** – Gateway to the Core Network.

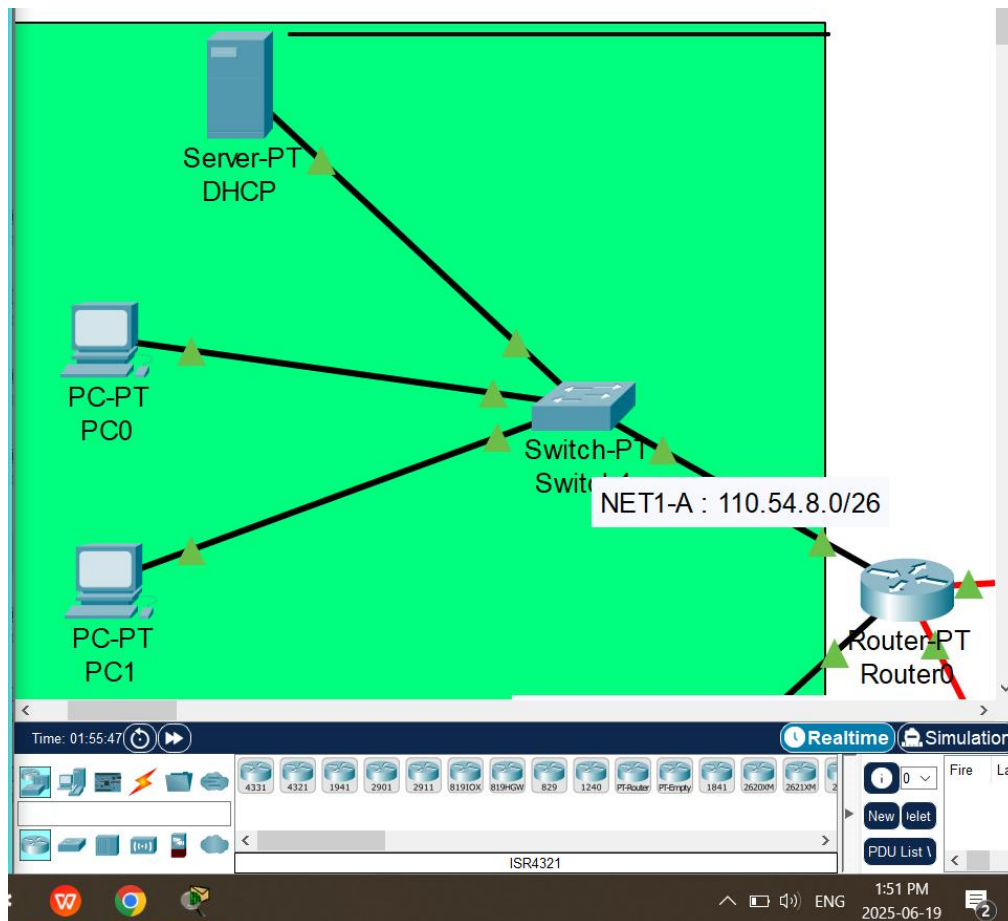


Image 14:NET1-A Connections

Configuration Summary:

Network Address: 110.54.8.0/26

Subnet Mask: 255.255.255.192

Default Gateway: 110.54.8.1

DHCP Server IP: Static IP inside subnet(110.54.8.2)

PC0 and PC1: Use DHCP for address assignment.

DHCP Server in NET1-A (University Network)

Purpose and Role

The DHCP Server in NET1-A is responsible for automatically assigning IP addresses and other network parameters to the wired devices (PC0, PC1, etc.). This automation eliminates the need for manual configuration and ensures efficient IP address management within the subnet.

It provides:

- IP address
- Subnet mask
- Default gateway
- DNS server

Configuration Details

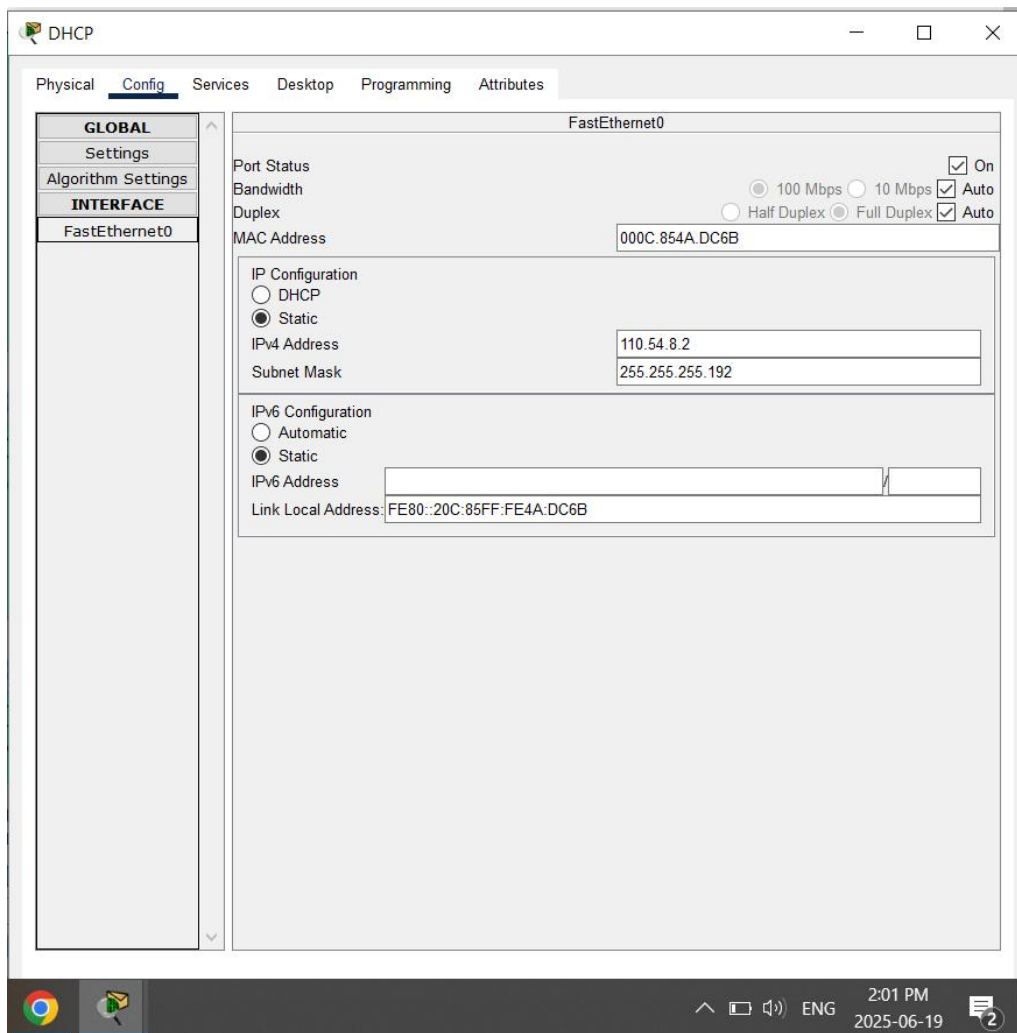


Image 15:DHCP server configurations

The DHCP server was given a static IP of 110.54.8.2/26 to ensure consistent network access. It's connected via FastEthernet0 and is responsible for assigning IPs to wired clients in NET1-A.

DHCP Setup

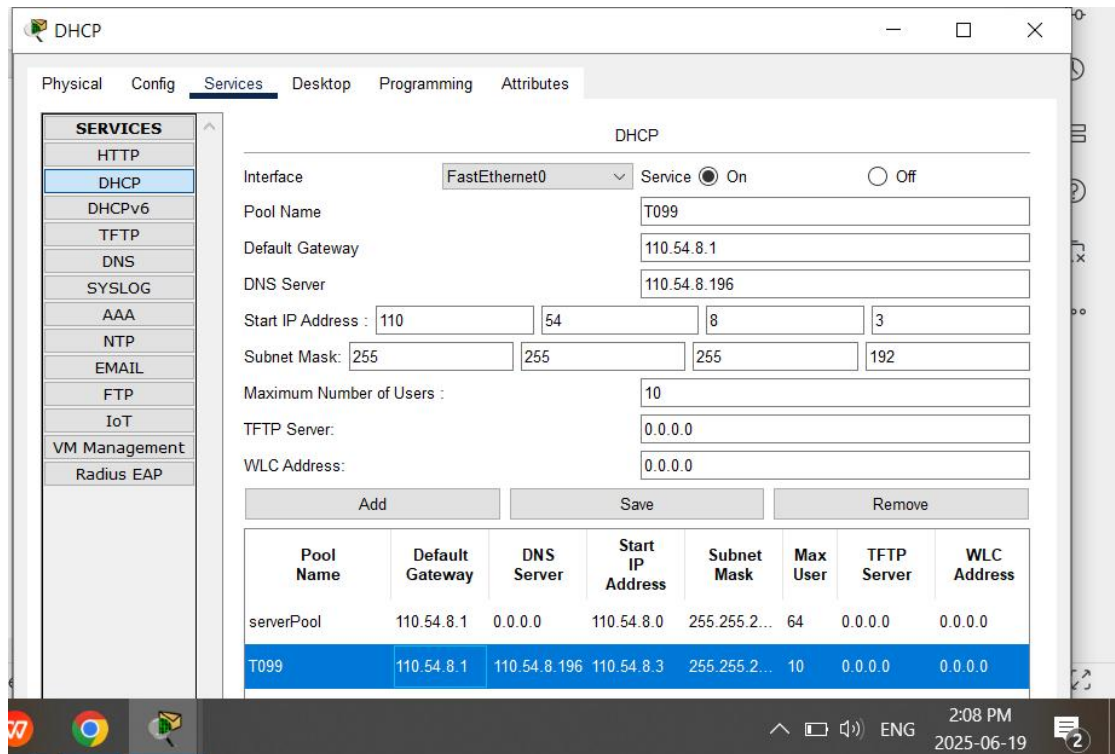


Image 16:DHCP setups pool T099

DHCP Server Pool Configuration T099 (NET1-A)

The DHCP server in NET1-A uses pool T099 to assign IPs from 110.54.8.3 with a /26 subnet mask. It sets the default gateway to 110.54.8.1 (Router0) and DNS to 110.54.8.196.

This setup ensures automatic IP configuration for up to 10 wired clients in NET1-A.

PCs Configuration – NET1-A

PC1 is configured to obtain its IP address automatically using DHCP.

Once connected to the switch in NET1-A, it receives:

IP Address: Assigned dynamically from the DHCP server (e.g., 110.54.8.3)

Default Gateway: 110.54.8.1 (Router0)

DNS Server: 110.54.8.196

This enables PC0 to communicate with other devices on the network and access services like the web or email via DNS.

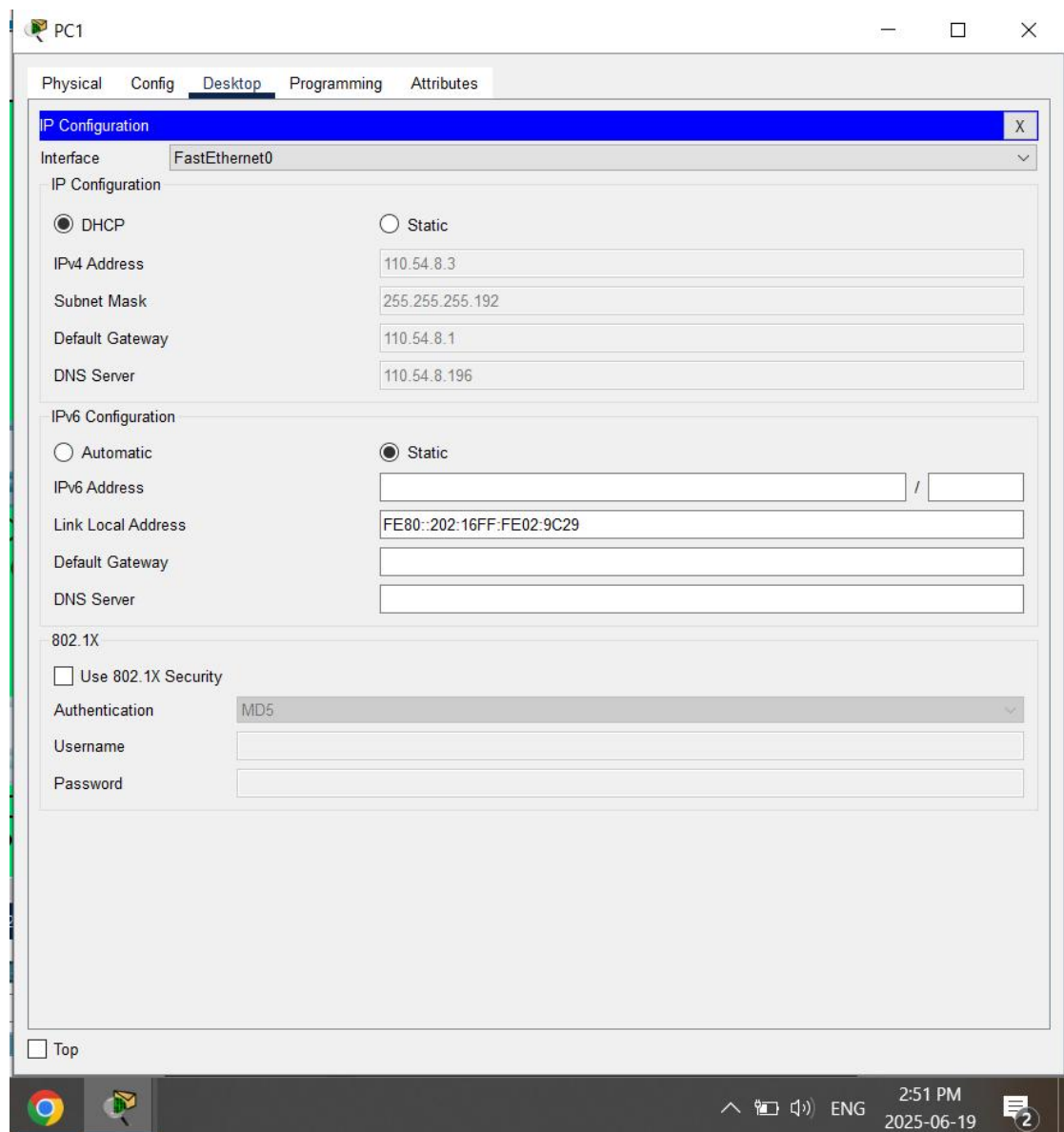


Image 17:PC dynamic IP address in NET1-A via DHCP server

University Network – NET1-B

1. Components

NET1-B is the wireless segment of the University Network, supporting mobile devices such as laptops and smartphones. It includes:

- **AccessPoint0** – Provides wireless access
- **Switch5** – Connects AccessPoint0 to Router0
- **Router0 (GE7/0)** – Connects NET1-B to the core and acts as the gateway
- **Server (in NET1-A)** – Provides DHCP and DNS services to both NET1-A and NET1-B
- **Wireless Devices** – Laptop0, Smartphone0, Tablet.

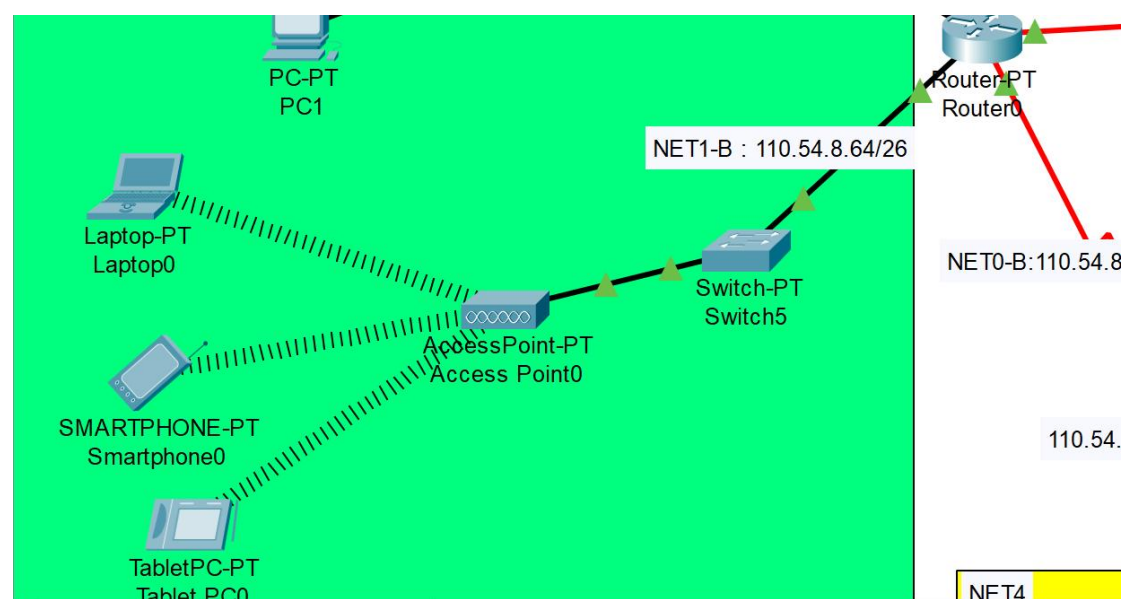


Image 18:NET1-B connections

2. IP Configuration

- **Network Address:** 110.54.8.64/26
- **Subnet Mask:** 255.255.255.192
- **Default Gateway:** 110.54.8.65 (Router0 interface)
- **DHCP Server:** 110.54.8.2 (Server in NET1-A)
- **Address Range:** 110.54.8.66 to 110.54.8.126 (62 usable hosts)

Router Configuration (DHCP Relay)

To enable DHCP for NET1-B from a remote server, the following command was configured on Router0:

```
Router(config)# interface [interface connected to NET1-B]
Router(config-if)# ip helper-address 110.54.8.2
```

This allows DHCP requests from wireless clients to reach the server in NET1-A.

3. DHCP Pools

Two pools were created on the central DHCP server:

Pool Name	Network	Default Gateway	Max Hosts
T099_pool2	110.54.8.64/26	110.54.8.65	62

Table 3:T099pool for NET2-B setup in DHCP server

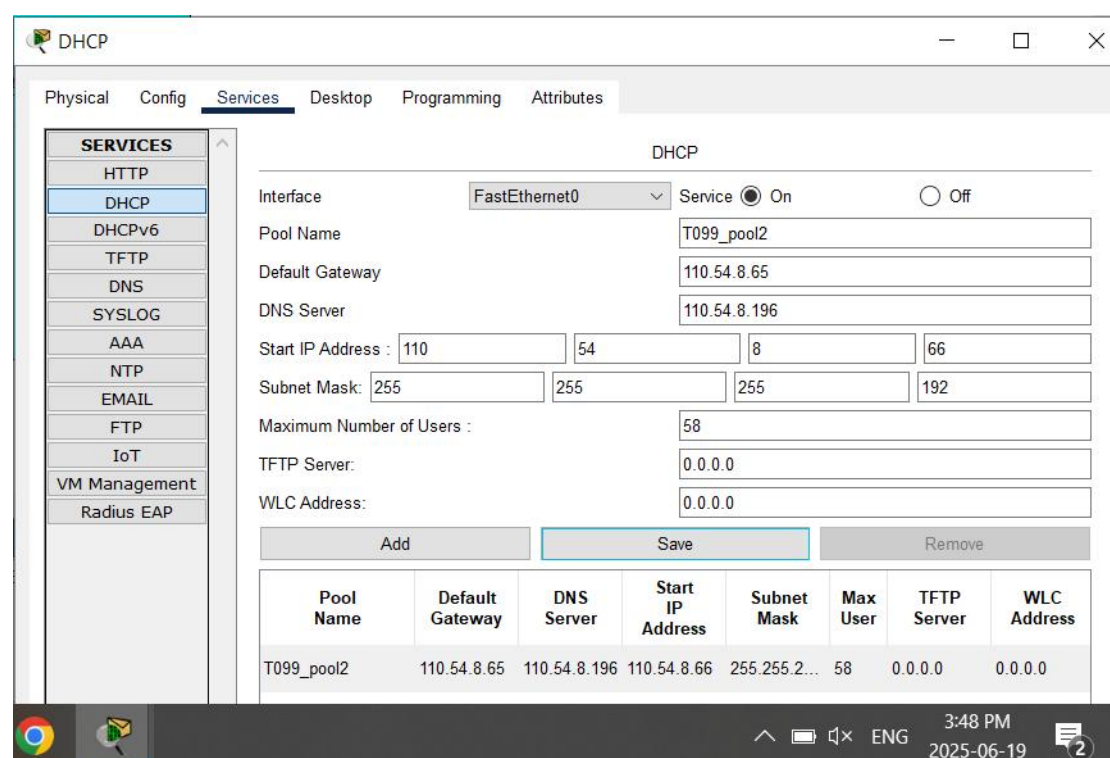


Image 19:T099_pool for NET2-B setup in DHCP server

4. Device IP Assignments in NET1-B

Device	Assigned IP	Status
--------	-------------	--------

Device	Assigned IP	Status
Laptop0	110.54.8.66	DHCP Success
Smartphone0	110.54.8.67	DHCP Success
Tablet	110.54.8.68	DHCP Success

Image 20:Dynammmice IP's assigned to end points in NET1-B

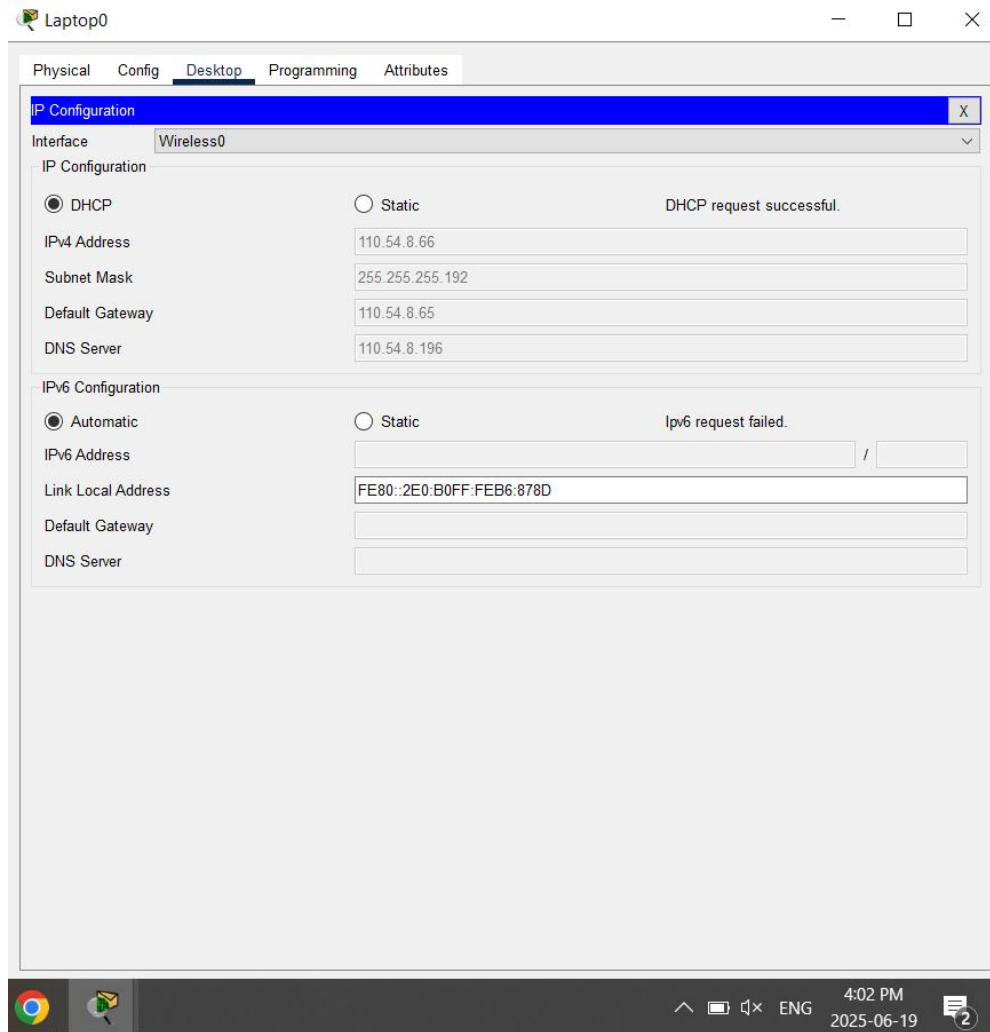


Image 21:HDPC IP for laptop in NET1-B

5. Troubleshooting Summary

Initially, wireless clients failed to obtain an IP address, defaulting to APIPA (169.254.x.x). After investigation, it was found that Router0 was not forwarding DHCP requests to the server. The issue was resolved by configuring:

```
ip helper-address 110.54.8.2
```

on the correct interface (GigabitEthernet7/0), which forwards DHCP requests to the server. Once this was set, all wireless clients successfully received IPs from **T099_pool2**.

6. Rationale for Using DHCP in NET1-B

DHCP was used in NET1-B to automate IP address assignment for wireless devices such as laptops and smartphones. These devices frequently connect and disconnect, making static IP configuration inefficient and error-prone. By using DHCP, IP addresses are assigned dynamically, reducing manual work, preventing address conflicts, and ensuring seamless connectivity for mobile users. This approach also allows centralized IP management, since both NET1-A and NET1-B share the same DHCP server.

7. Access Point Configuration – NET1-B

The Access Point in NET1-B allows wireless devices like laptops, smartphones, and tablets to connect to the network. We set the SSID to ENC3320T099, which is the wireless network name. For security, we used WPA2-PSK (a strong encryption method) with the password Home_T099. This ensures that only authorized users can join the network.

The coverage range was set to 140 meters to ensure all wireless devices in the university area can stay connected. The Access Point works with the DHCP server to give each connected device an IP address automatically.

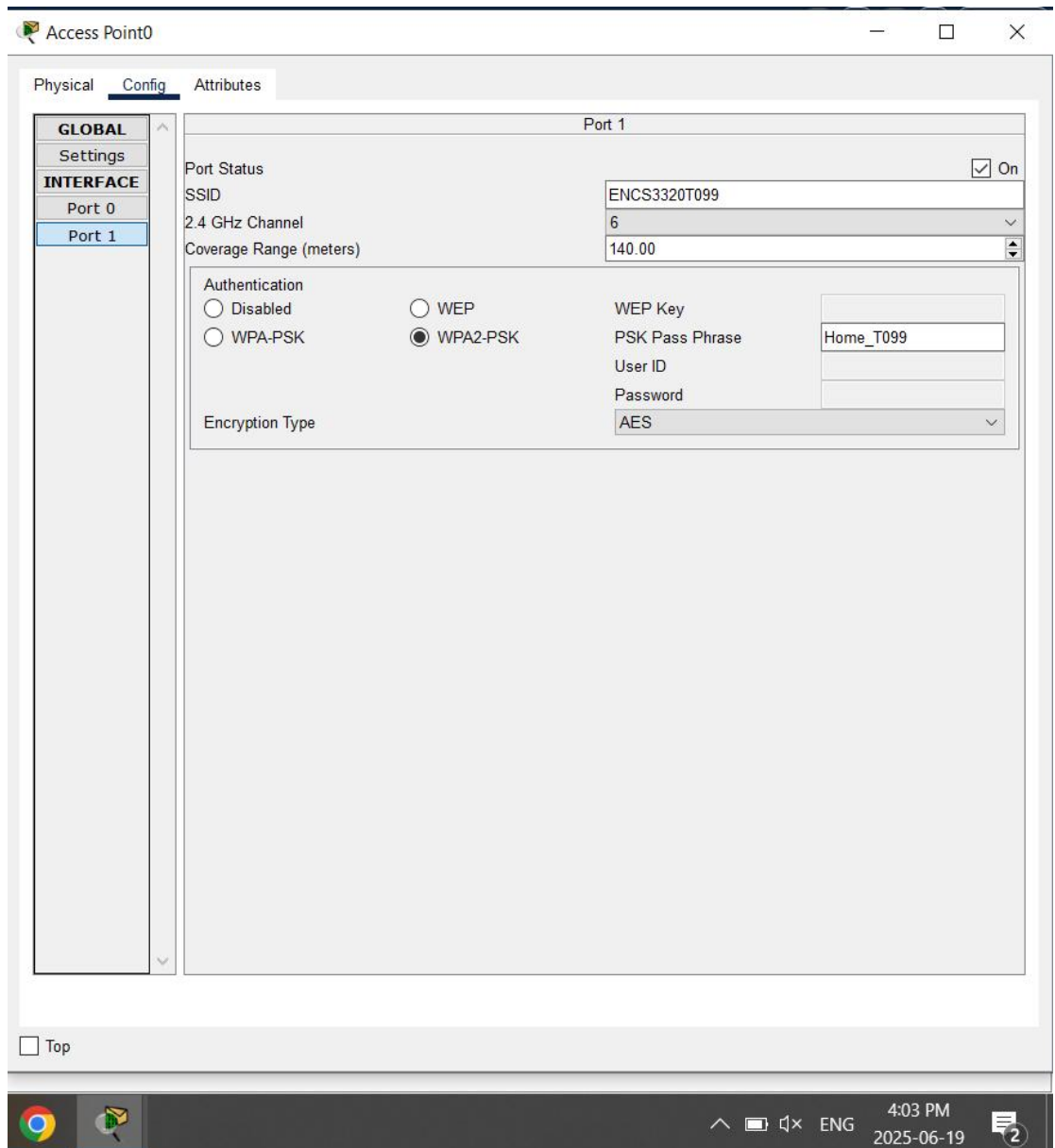


Image 22:Access point setup

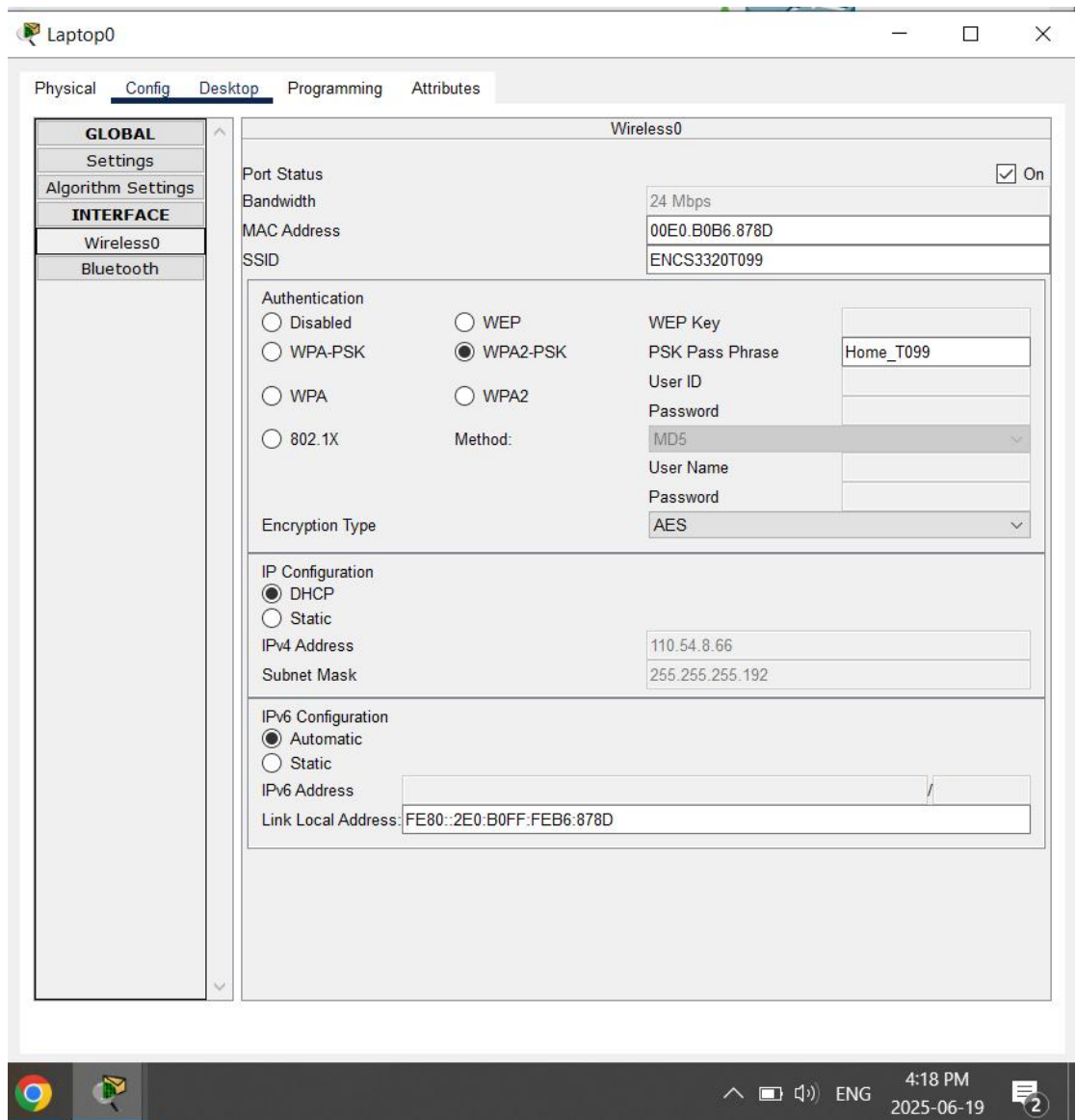


Image 23:Connect to Access point via Laptop setup

NET2 – Street Network

Subnet: 110.54.8.128/27

The Street Network connects the Central Office Server, a Cell Tower, and several Smartphones to Router1. To use IP addresses efficiently and maintain clarity in design, this network was divided into two smaller subnets.

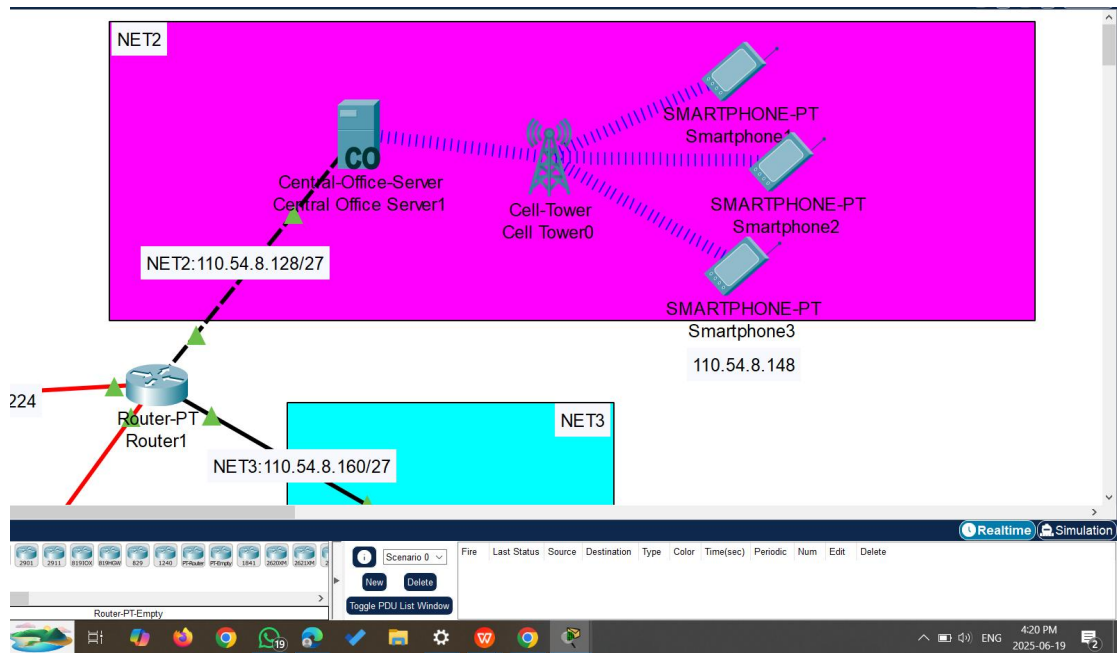


Image 24:NET2 – Street Network connections

Subnetting and Design Rationale

The original assigned block for NET2 was 110.54.8.128/27, providing 32 IP addresses (30 usable). However, a technical limitation arose when configuring the Central Office Server, which uses two interfaces:

- One for the backbone connection (to Router1)
- One for the cellular 3G/4G connection (to the Cell Tower)

The server does not allow assigning two IPs from the same subnet to different interfaces. This restriction made it necessary to split NET2 into two smaller subnets using a /28 mask (each with 16 total addresses, 14 usable).

Subnet Allocation

Subnet 1: 110.54.8.128/28

Assigned to the **backbone interface** of the Central Office Server and Router1

- IP Range: 110.54.8.129 – 110.54.8.142
- Used to establish the wired link to Router1

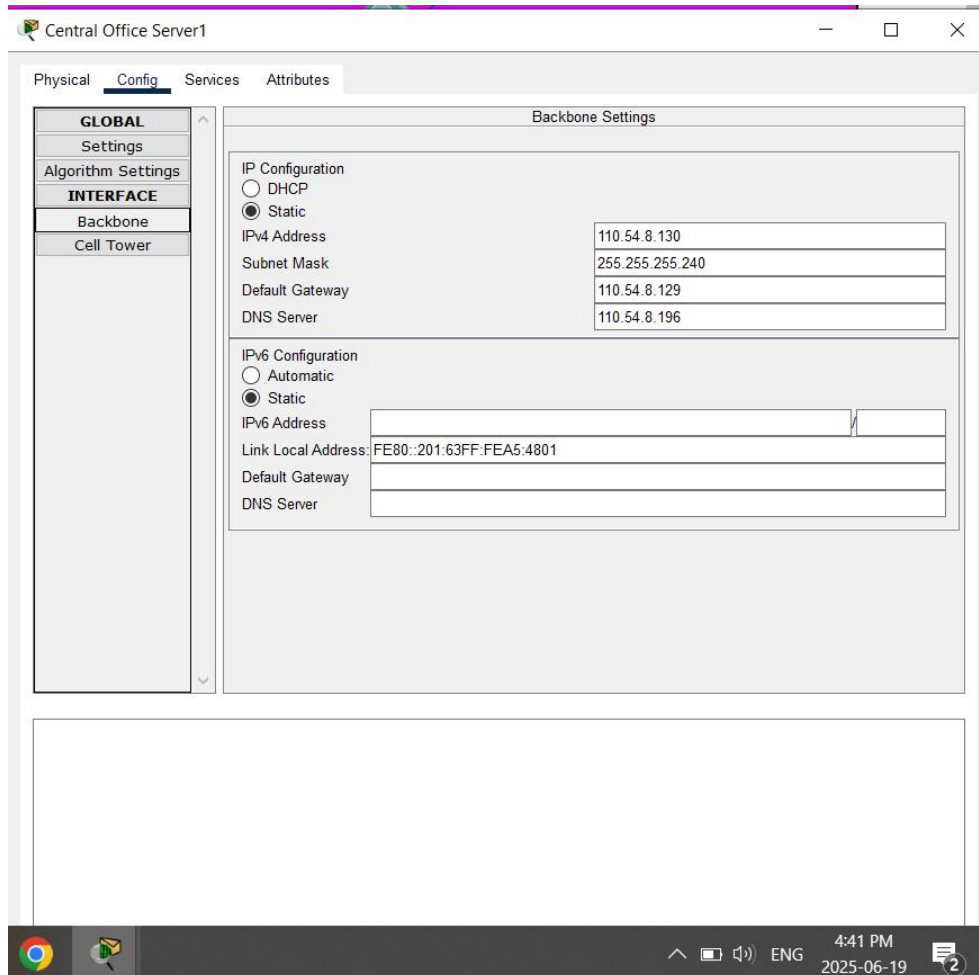


Image 25: CO server backbone interface setup

Subnet 2: 110.54.8.144/28

Assigned to the 3G/4G interface of the Central Office Server and the Cell Tower

- IP Range: 110.54.8.145 – 110.54.8.158
- Used for wireless communication with smartphones

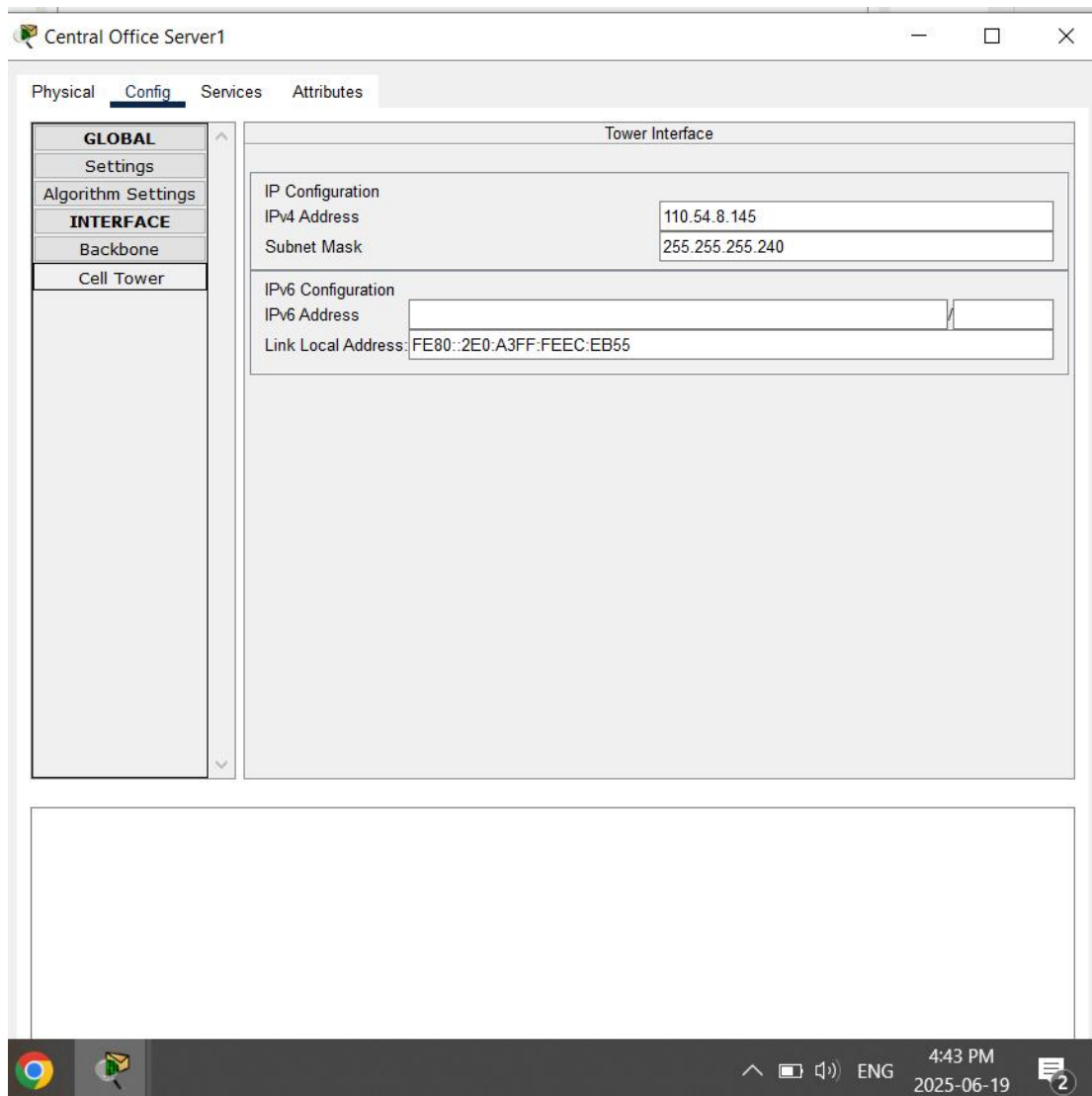


Image 26:CO server cell Tower interface setup

This design ensured proper IP assignment without violating interface restrictions, and logically separated internal and external communication.

Cell Tower Configuration Overview:

The Cell Tower0 acts as a wireless provider in the Street Network (NET2), allowing smartphones to connect through a 3G/4G interface. It is configured with the provider name “T099”, and the “Allow External Access” option is turned ON, which lets smartphones access services outside the local network—such as web or email—just like how real-world mobile networks work. The tower connects to the Central

Office Server, which provides DHCP service to assign IP addresses to smartphones automatically.

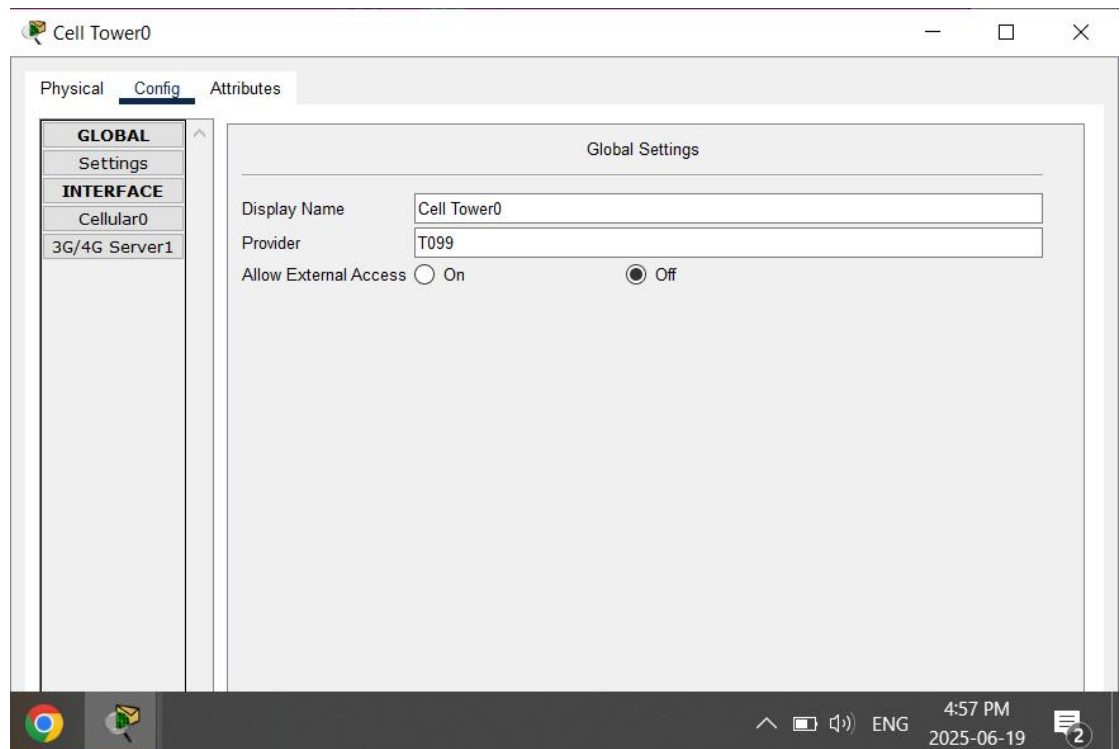


Image 27:Cell tower provider name setup

DHCP Configuration for Smartphones in NET2

Although smartphones in the Street Network (NET2) were initially configured with static IPs, the Central Office Server (IP: 110.54.8.145) includes a properly configured DHCP service for the subnet 110.54.8.144/28. This DHCP server is capable of dynamically assigning IPs to smartphones connected via the Cell Tower.

The DHCP setup includes:

- **Start IP Address:** 110.54.8.145
- **Maximum Users:** 13

- **IP Address Range:** 110.54.8.145 to 110.54.8.157
- **DNS Server:** 110.54.8.196 (in the Datacenter)

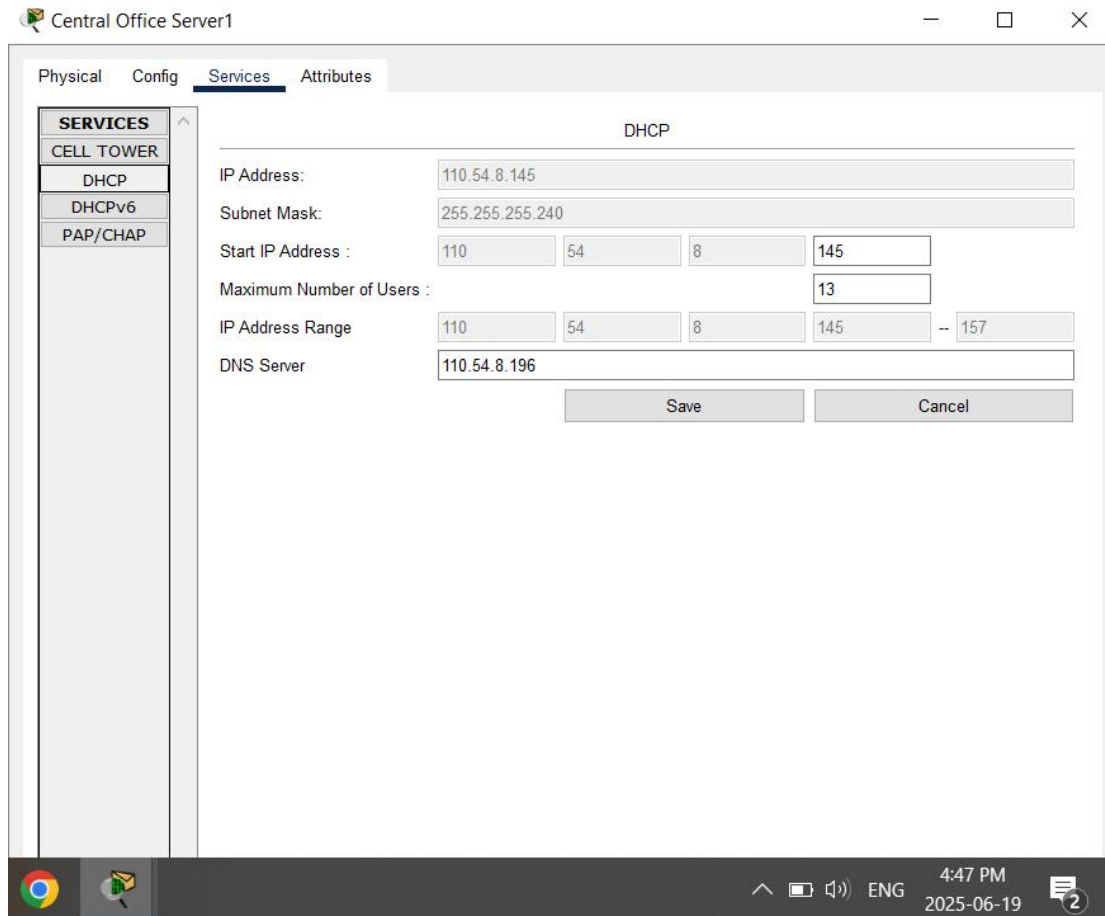


Image 28:Cell tower DHCP setup

This setup allows smartphones to receive IPs automatically without manual input, making the network more flexible and easier to manage.

In our final setup, DHCP was enabled and functional, so smartphones can either use dynamic IPs (if configured with DHCP) or static IPs (for testing/stability during simulation).

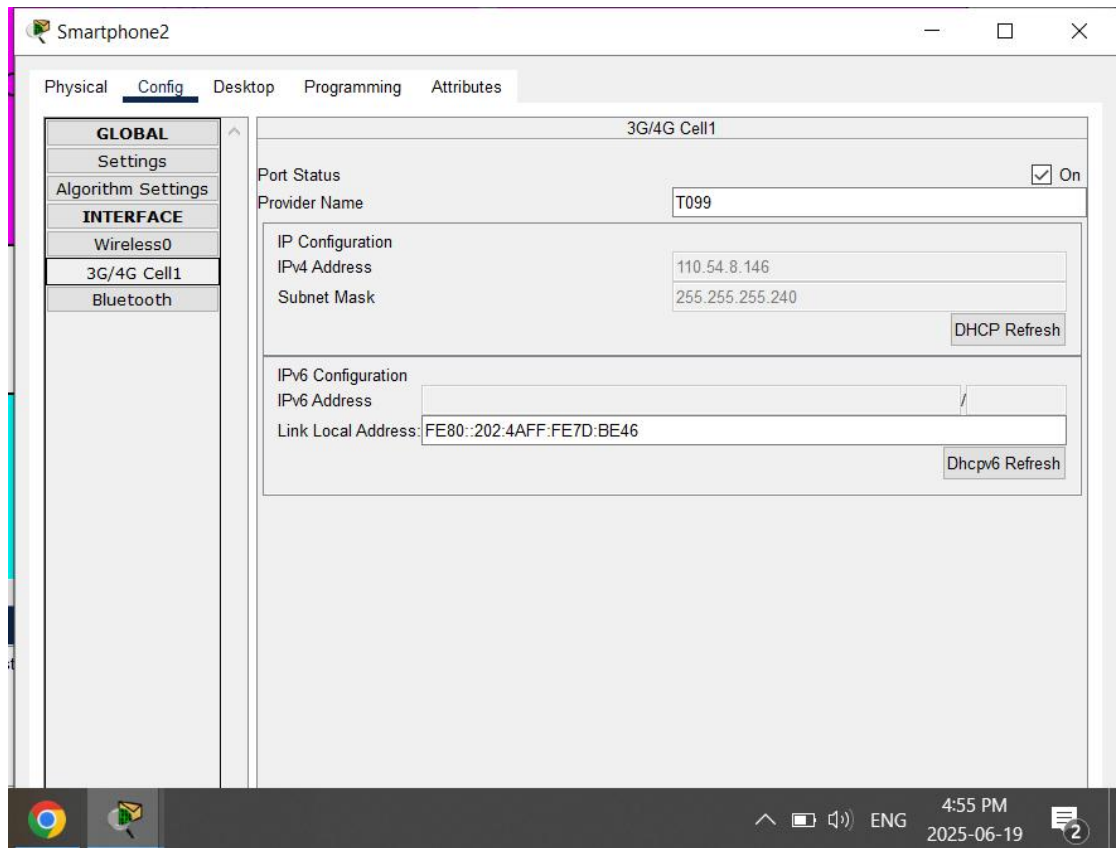


Image 29:smartphone configuration setup

NET3 – Home Network

Overview:

NET3 represents the Home Network segment in our topology. It uses the subnet 110.54.8.160/27, which provides up to 30 usable IP addresses — more than enough for the few home devices involved. This network includes two PCs (PC2 and PC3) connected to a switch (Switch7), which is in turn connected to Router1.

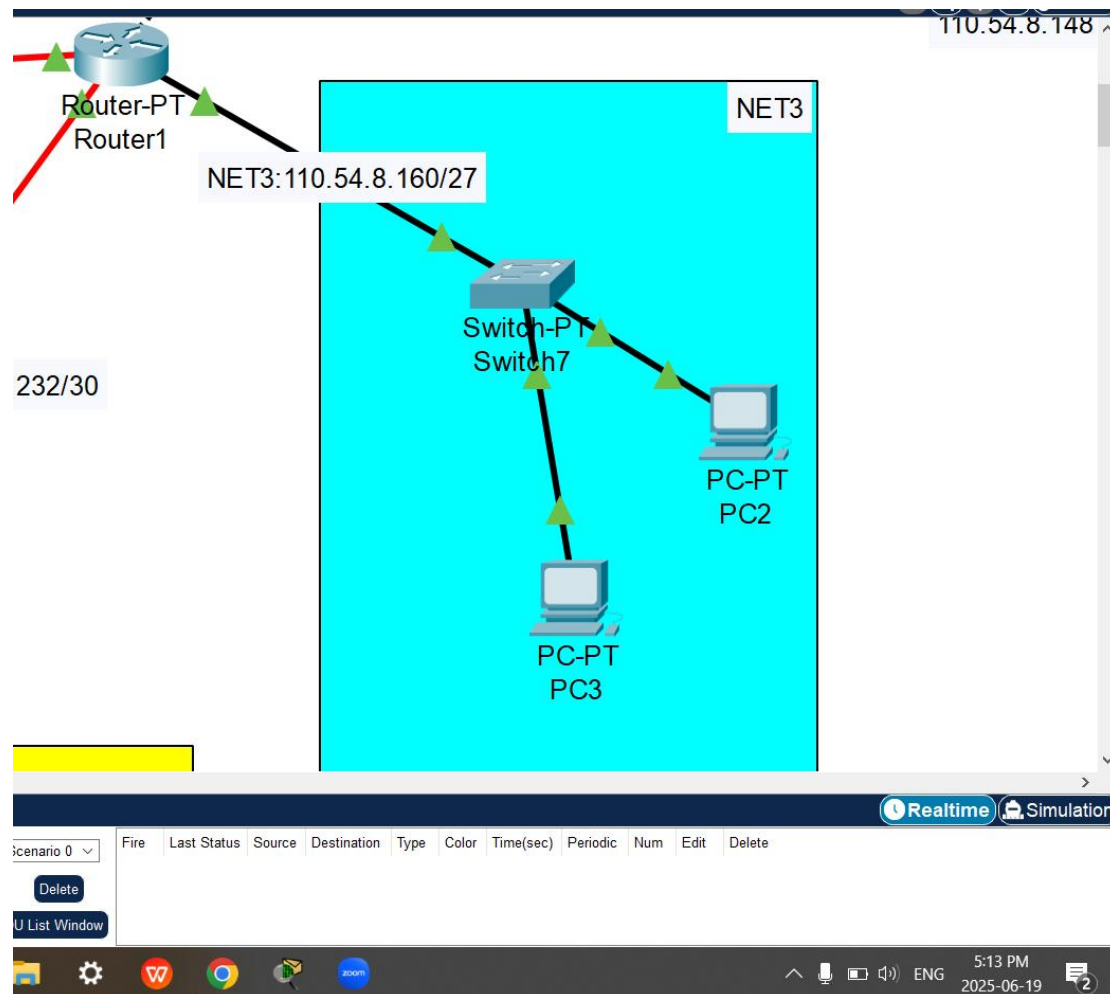


Image 30:NET3 Home - connections

Configuration Details:

- **IP Addressing:** Both PC2 and PC3 were assigned static IP addresses within the range of 110.54.8.161 – 110.54.8.190.
- **Subnet Mask:** 255.255.255.224

- **Default Gateway:** Set to the IP address of Router1's interface connected to NET3 (110.54.8.161).
- **Switch:** No configuration was needed on Switch7 beyond cabling, as it operates in Layer 2 (forwarding frames within the local network).

We used static IPs in NET3 because the home PCs are fixed devices that don't move around, and the network is small and stable. This makes static addressing simpler and more efficient than setting up DHCP.

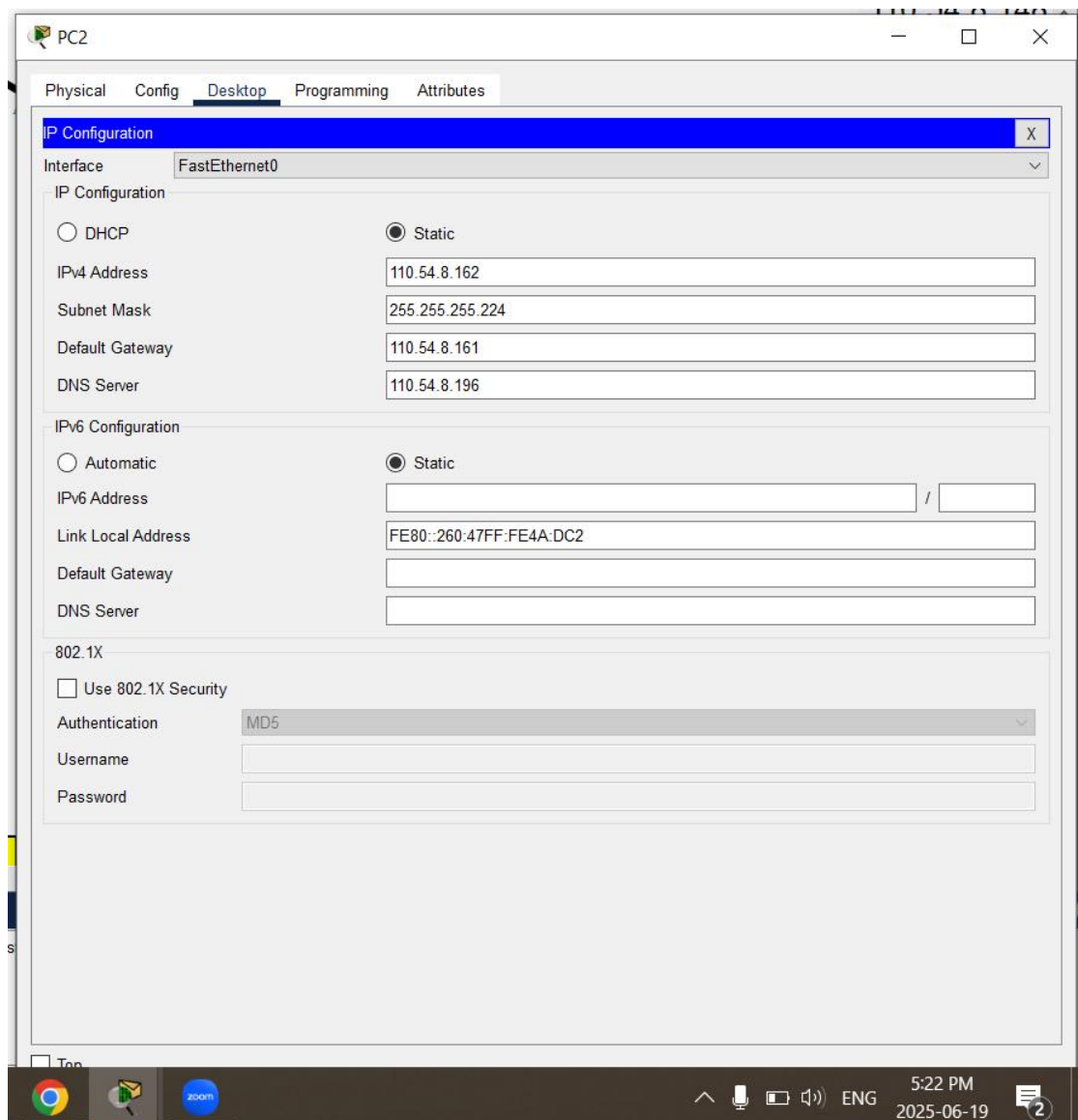


Image 31: Home pc static config setup

Usage Purpose:

This home network simulates typical home PCs used to access university resources (like the DNS, Web, or Email servers located in NET4). It tests basic connectivity, routing, and access to services provided across the whole project.

NET4 – Datacenter Network (110.54.8.192/27)

The Datacenter network hosts three critical servers: Web Server, Mail Server, and DNS Server, all of which are directly connected to Switch6, which is in turn connected to Router2. This segment of the network uses the subnet 110.54.8.192/27, which provides up to 30 usable IP addresses – more than enough for the servers and future scalability.

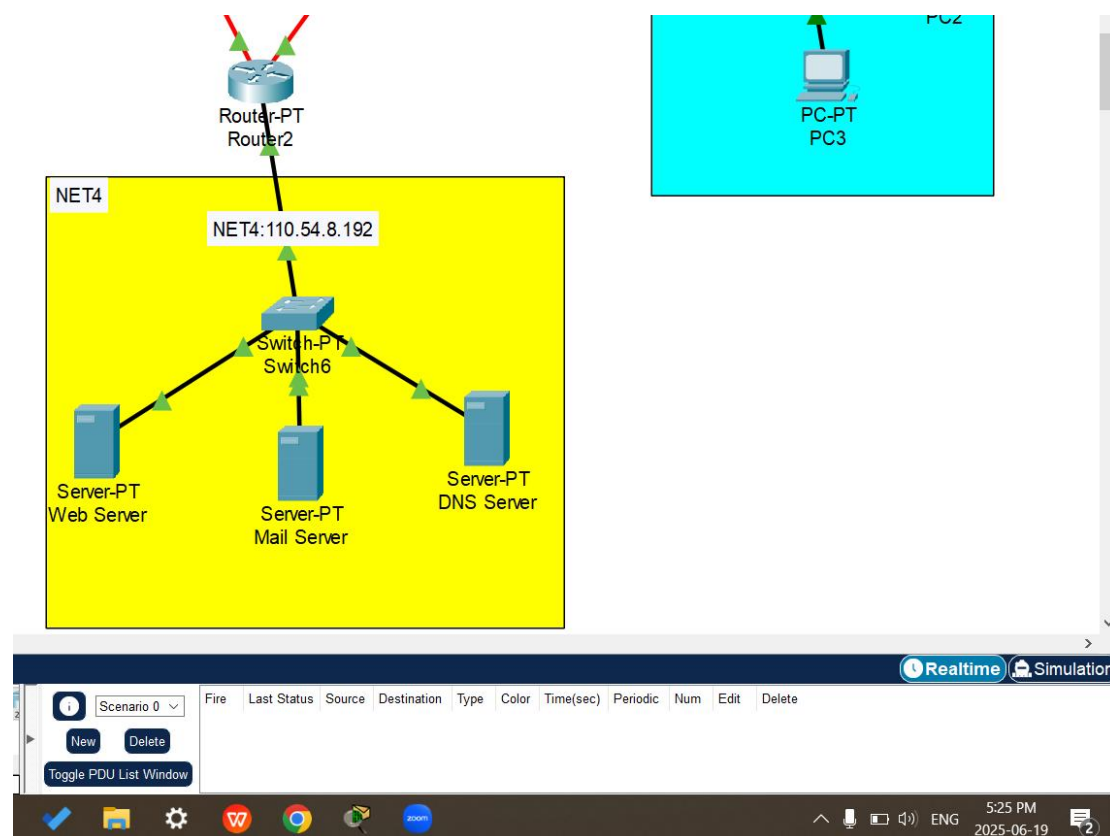


Image 32:NET4 Datacenter-Connections

IP Configuration

Each server in NET4 was manually assigned a **static IP address** from the subnet range. This was done because:

- Servers must always be reachable at a fixed address.
- Services like DNS, Web, and Email need consistent IPs for proper domain resolution and routing.

IP Assignment:

- Web Server: 110.54.8.194.
- Mail Server: 110.54.8.195.
- DNS Server: 110.54.8.196.
- Default Gateway (Router2): 110.54.8.222 (via /30 point-to-point subnet).

1. Web Server Configuration

The Web Server is located in the Datacenter network (NET4: 110.54.8.192/27) and is connected to Switch6, which connects it to Router2 for external communication. This server is responsible for hosting and serving the main website of the university (e.g., www.coe-bzu.com), and it must always be available with a fixed identity. Therefore, it is configured with a static IP address.

IP Address: 110.54.8.194

Subnet Mask: 255.255.255.224

Gateway: 110.54.8.193

Services Enabled: HTTP (Port 80)

Static IP addressing is used because the Web Server is a critical infrastructure component. Static IPs ensure stability, make DNS mapping possible, and help avoid connectivity issues caused by IP changes.

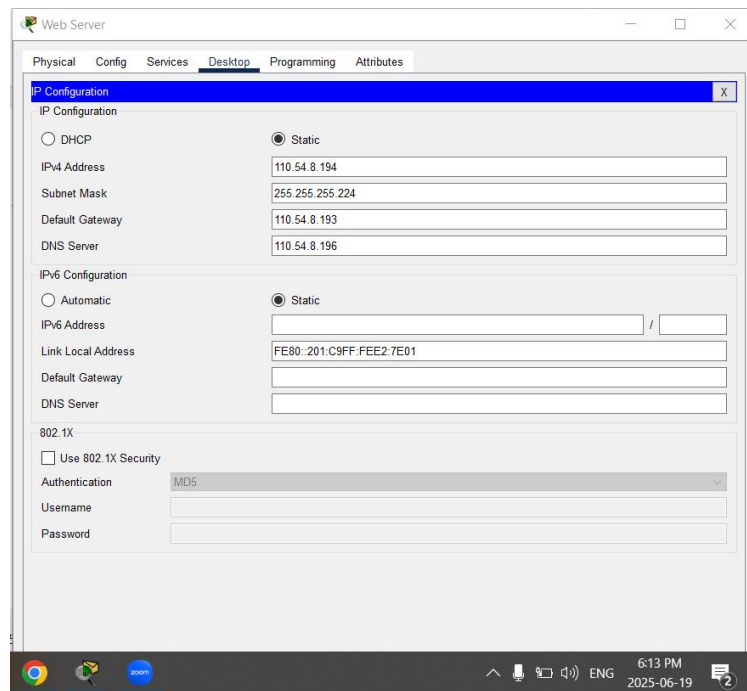


Image 33: Web server configuration

Service Activation:

- In the **Services** tab, the **HTTP** and **HTTPS** services were enabled to allow users to access the website securely.
- No username/password authentication was required, as the server hosts a public faculty webpage.

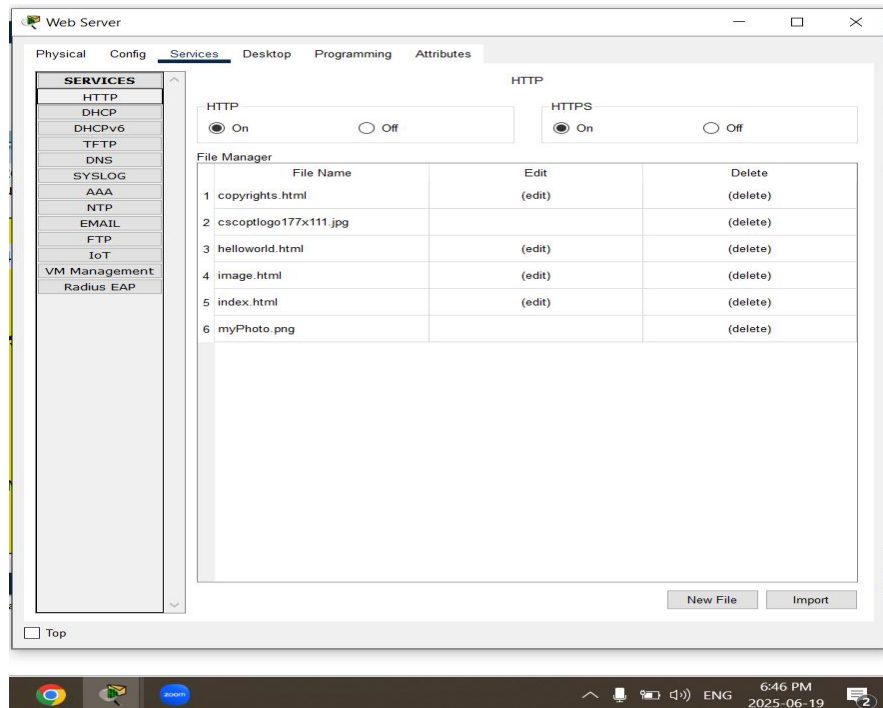


Image 34:WEB server activation

Web Page Creation (index.html):

The index.html file was created to showcase a customized, visually appealing web page titled "Faculty of Engineering and Technology", representing the College of Engineering at Birzeit University.

- **Tab Title:** COE-Birzeit
- **Main Heading:** Faculty of Engineering and Technology
- **Description:** A detailed paragraph introducing the faculty's mission and values.
- **Team Information:** Since this project was done individually, only one team member is listed: **Rawand Radi – ID: 1221054**
- **Project Summary:** A paragraph explains the purpose of the project and how it simulates real-world enterprise networks using Cisco Packet Tracer.
- **Gallery Section:** Includes embedded images like the COE logo and a personal photo titled myPhoto.png to humanize and personalize the page.

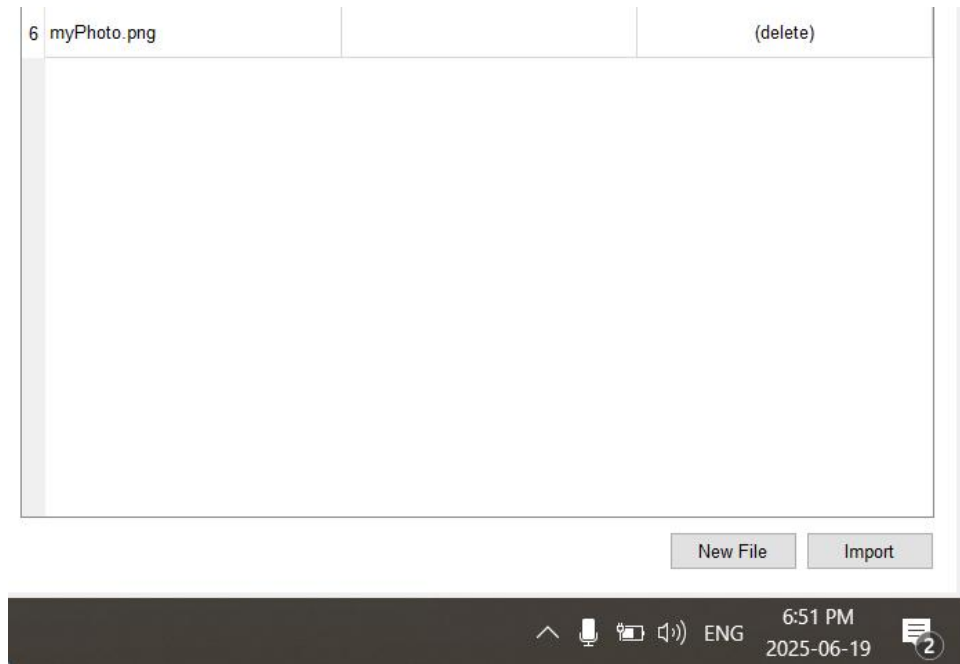


Image 35:MyPhoto uploade

- **Quick Links:** Hyperlinked pages like helloworld.html, image.html, and more for basic navigation and functionality.

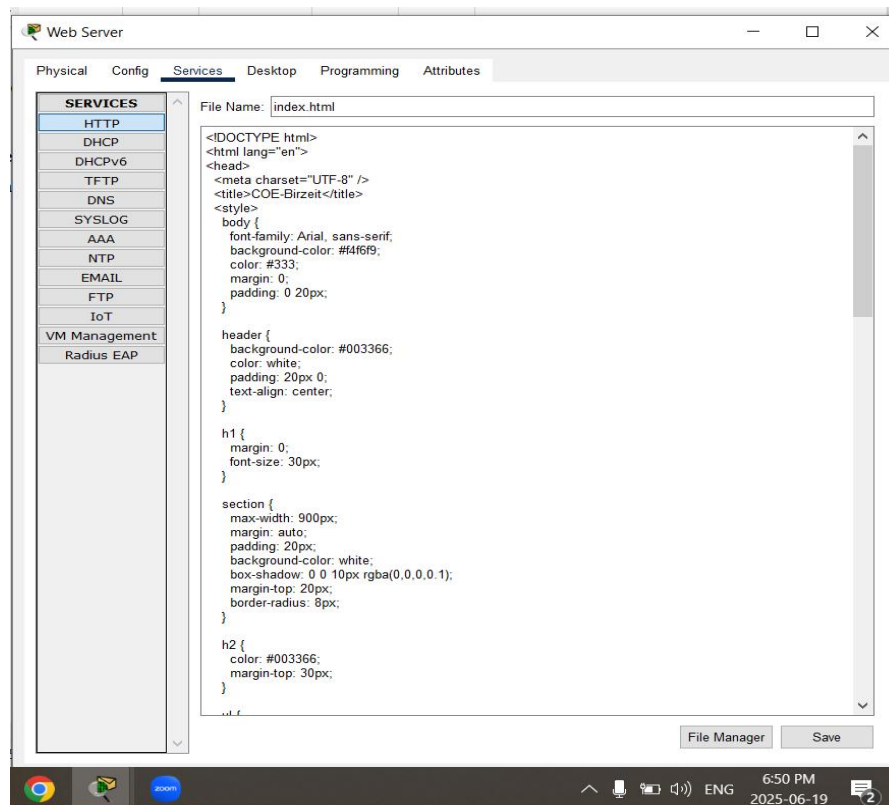


Image 36:Index.html setup

Why This Setup

Hosting Locally: The Web Server hosts the university and project info internally. This simulates how institutions manage their web content without relying on external servers. Secure & Controlled Environment: Static IP ensures reliable access. HTTPS enhances security.

2. Mail Server Configuration

The Mail Server located in NET4 (Datacenter Network) was configured to provide both sending and receiving email functionalities. It was assigned the hostname mail.coe.birzeit.edu and given a static IP address of 110.54.8.195.

IP Configuration

The following static IPv4 settings were applied:

- **IP Address:** 110.54.8.195
- **Subnet Mask:** 255.255.255.224 (/27)
- **Default Gateway:** 110.54.8.193
- **DNS Server:** 110.54.8.196

A static IP was chosen because servers are required to remain reachable at consistent, predictable addresses. Dynamic addressing was avoided to prevent DNS resolution issues.

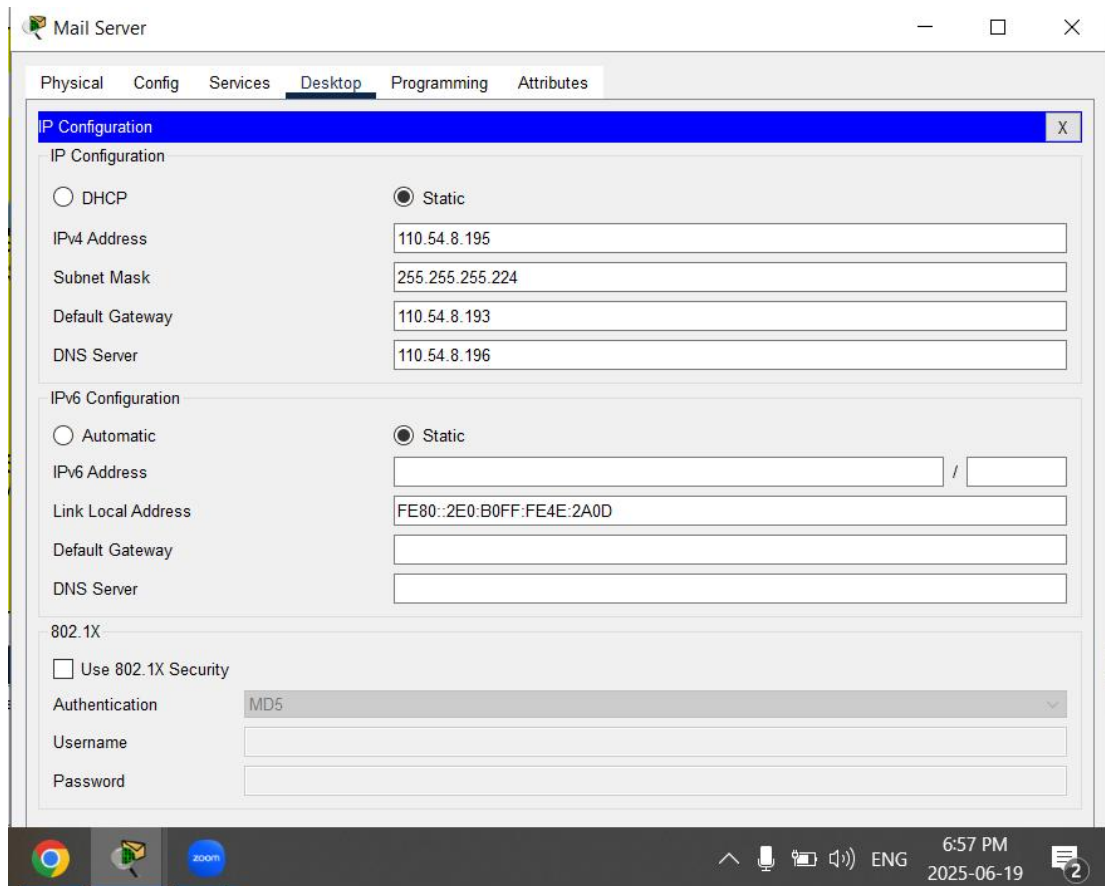


Image 37: Mail server configuration

Mail Services

To support email communication, only the necessary mail protocols were enabled:

- **SMTP** was activated for sending emails.
- **POP3** was enabled to allow users to retrieve incoming emails.

Other services, such as IMAP, were left disabled to reduce system complexity and improve security.

Domain & Hostname Configuration

The server was assigned the domain name `coe.birzeit.edu`, with the full mail hostname defined as `mail.coe.birzeit.edu`. A DNS record was created and configured on the DNS Server to ensure this domain name correctly resolves to 110.54.8.195.

User Accounts Setup

Three user accounts were created—each representing a different network (Home, Street, and University). The usernames were constructed using the format {NetworkName}{SID}, and the student ID 1221054 was used for all account credentials.

Username	Password	Email Address	Network
Home1221054	1221054	Home1221054@coe.birzeit.edu	Home (NET3)
Street1221054	1221054	Street1221054@coe.birzeit.edu	Street (NET2)
University1221054	1221054	University1221054@coe.birzeit.edu	University (NET1)

Table 4:Mail server User Accounts Setup table

These credentials were used to authenticate email clients for testing SMTP and POP3 communication within the network.

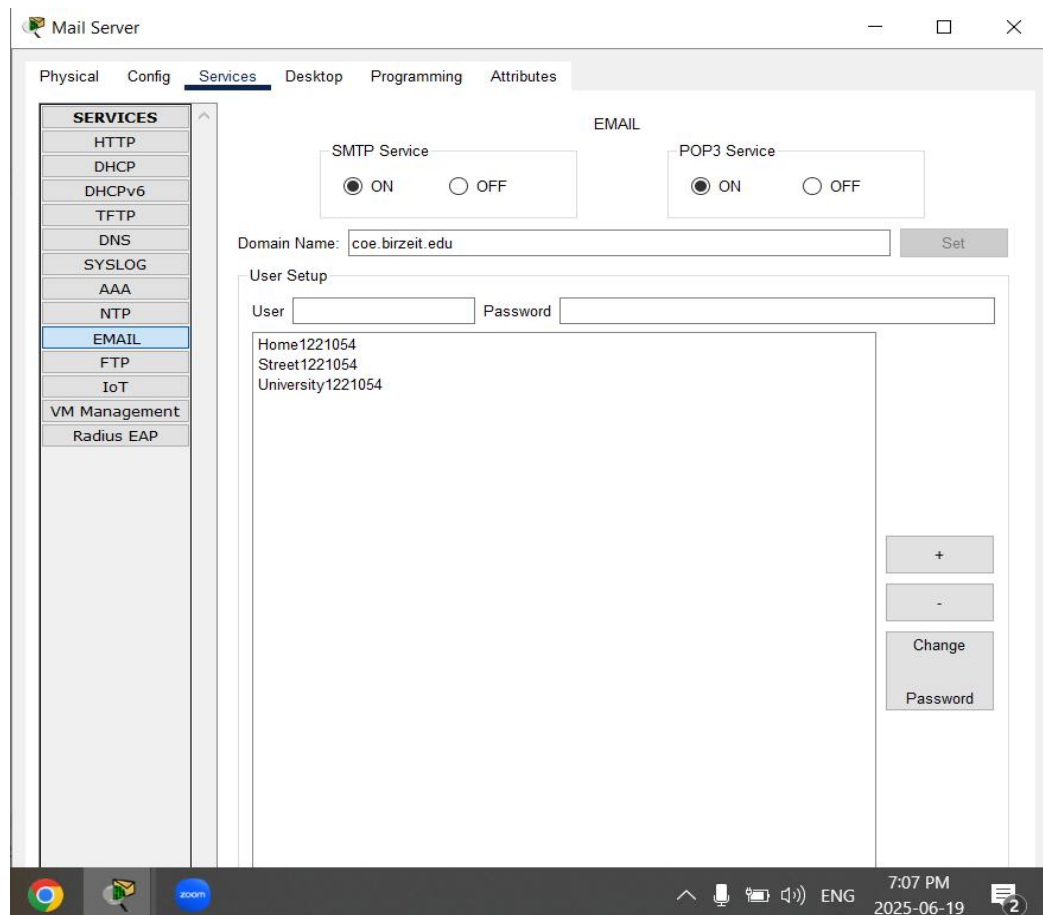


Image 38: Mail server setup and User Accounts Setup

DNS Server Configuration

The DNS Server was configured within NET4 (Datacenter Network) to provide essential domain name resolution services across the entire network. A static IP configuration was used to ensure reliable and uninterrupted DNS availability.

IP Configuration

The DNS Server was manually configured with the following network settings:

- **IP Address:** 110.54.8.196
- **Subnet Mask:** 255.255.255.224 (/27)
- **Default Gateway:** 110.54.8.193
- **DNS Server:** (self-reference) 110.54.8.196

A static IP was required for the DNS server to maintain consistent reachability from all other network devices, ensuring that domain queries are always resolvable to the correct IP addresses.

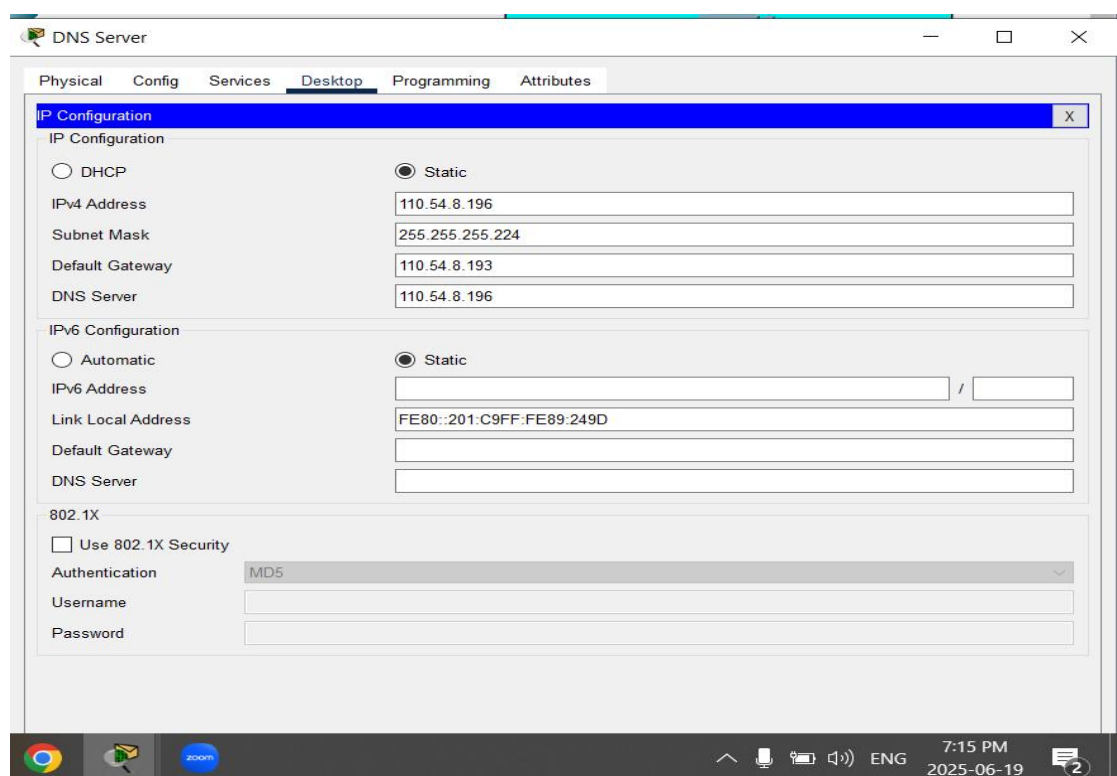


Image 39:DNS server configuration

DNS Service Activation

The DNS service was enabled under the Services > DNS tab. Multiple A (Address) records were manually created to map domain names to the IP addresses of internal services. This allowed devices across different subnets to resolve service hostnames without relying on external DNS infrastructure.

Resource Records Created

Three entries were added to the DNS server's database:

No.	Name	Record Type	Mapped Address
0	coe.birzeit.edu	CNAME	mail.coe.birzeit.edu
1	mail.coe.birzeit.edu	A Record	110.54.8.195 (Mail Server)
2	www.coe.birzeit.edu	A Record	110.54.8.194 (Web Server)

Table 5:Resource Records table

- The CNAME record for coe.birzeit.edu was linked to mail.coe.birzeit.edu, allowing aliasing of mail traffic.
- A Records were used for direct IP mappings to both mail and web servers, enabling domain-based access to the services.

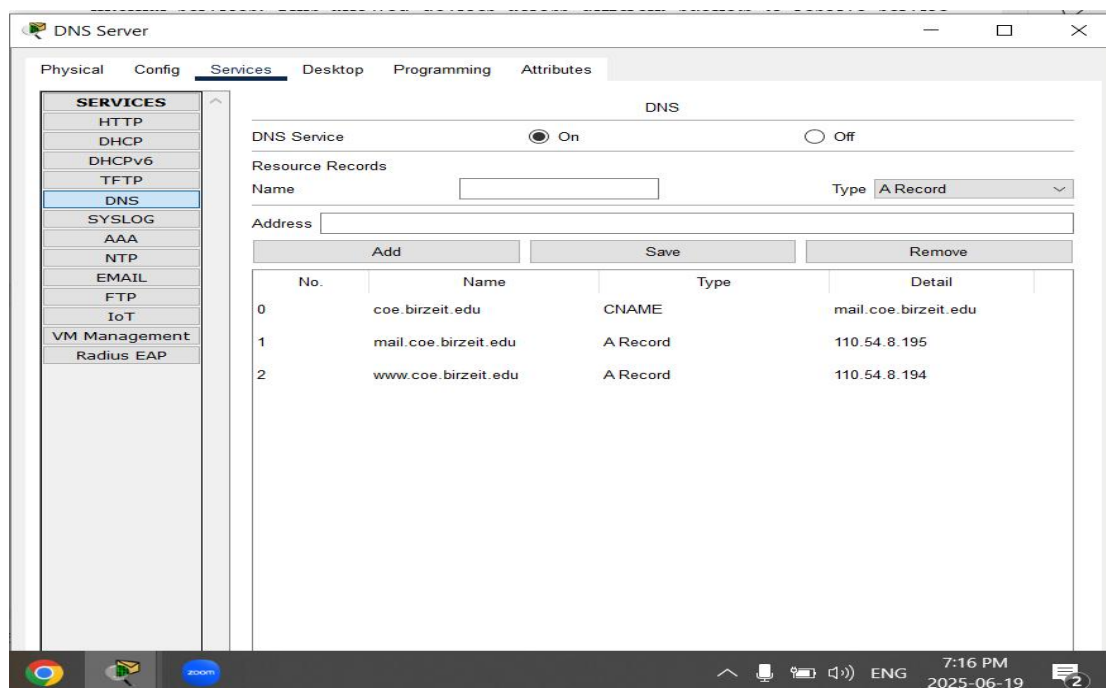


Image 40:DNS server setup

Purpose & Impact

The DNS Server ensured the following:

- All web traffic to `www.coe.birzeit.edu` was routed to the Web Server.
- All email applications could resolve `mail.coe.birzeit.edu` for SMTP and POP3 communication.
- Alias redirection via `coe.birzeit.edu` improved domain organization.

Without this DNS setup, users would have been required to use IP addresses, reducing usability and increasing configuration complexity.

Why DNS Was Important

Without the DNS server, devices would need to remember and use IP addresses manually, which is impractical in large or service-diverse networks. The DNS server served as a central resolver, enabling communication between mail clients, web browsers, and networked systems using human-readable domain names

Results and discussion

This section presents the verification and validation process of the implemented network design. All tasks have been tested in Cisco Packet Tracer and supported with screenshots that display the system's date and time as evidence of functionality.

Labeled Topology

Description:

The final topology has been fully labeled with subnet names, router and switch identifiers, and device roles. This helps in visual clarity and network structure understanding.

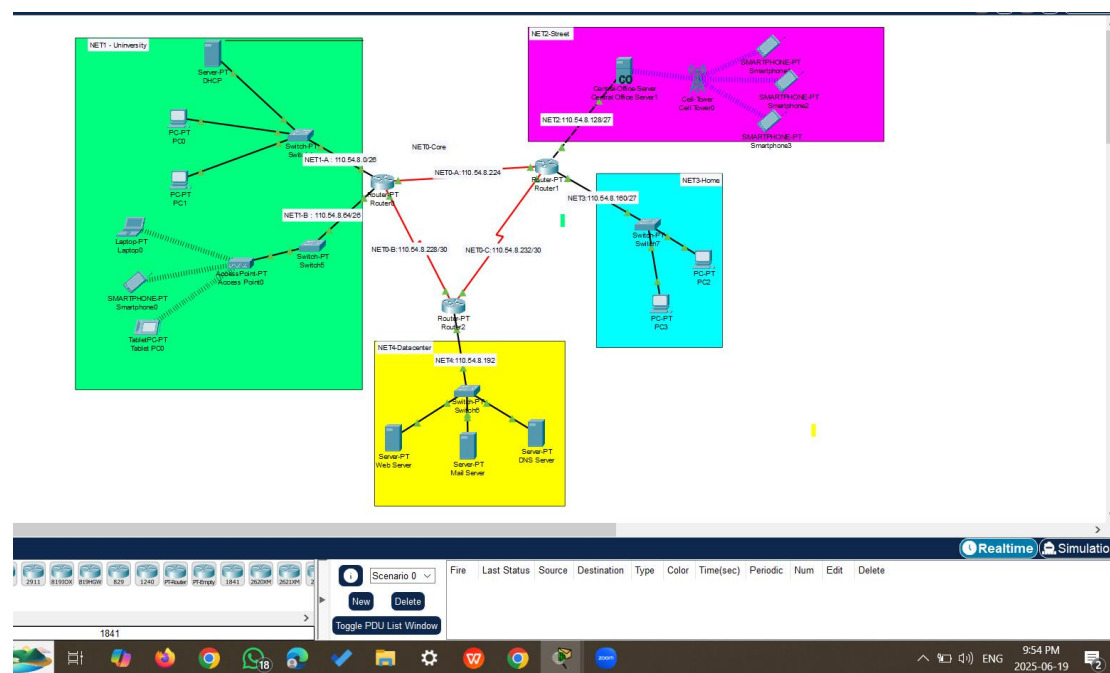


Image 41:Labeled Topology

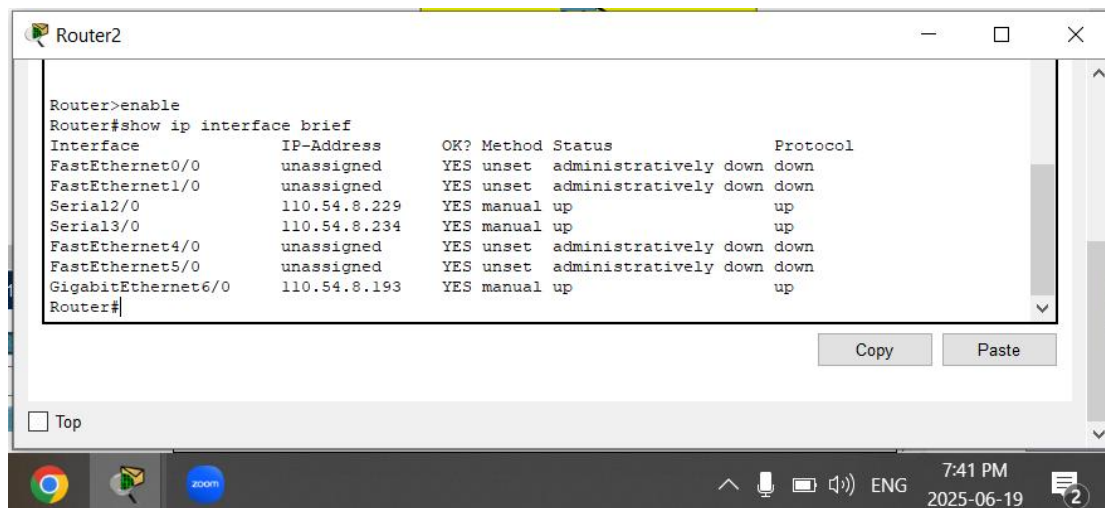
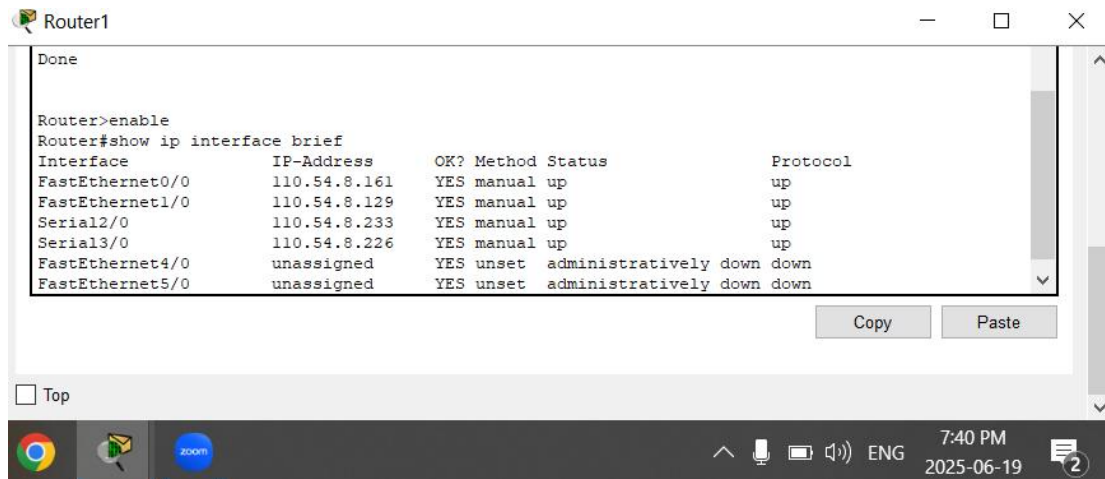
1. Static IP Configuration for Routers

Each router was configured manually with static IP addresses for each interface using the ip address command. This ensures consistent addressing across subnetworks.

```

show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset   administratively down down
FastEthernet1/0 unassigned      YES unset   administratively down down
Serial2/0      110.54.8.230    YES manual  up          up
Serial3/0      110.54.8.225    YES manual  up          up
FastEthernet4/0 unassigned      YES unset   administratively down down
FastEthernet5/0 unassigned      YES unset   administratively down down
GigabitEthernet6/0 110.54.8.1     YES manual  up          up
GigabitEthernet7/0 110.54.8.65     YES manual  up          up
Router#

```



Each router interface was configured manually with static IP addresses based on the subnetting plan. This guarantees stable communication paths and proper OSPF area assignments. The interfaces connecting to other routers use /30 subnets, while those connected to LANs use appropriate /27 or /26 subnets.

Dynamic IP Configuration for End Devices

End devices in networks where DHCP is enabled (e.g., NET1-B and Street network smartphones) successfully received IP addresses dynamically.

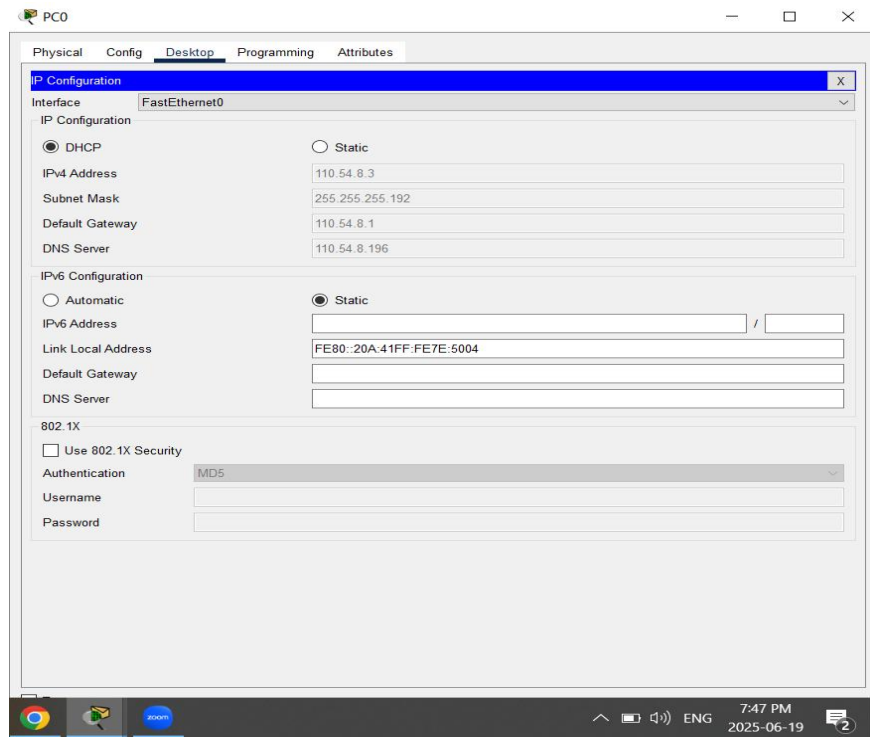


Image 42:Dynamic IP Configuration for PC0(In university area)

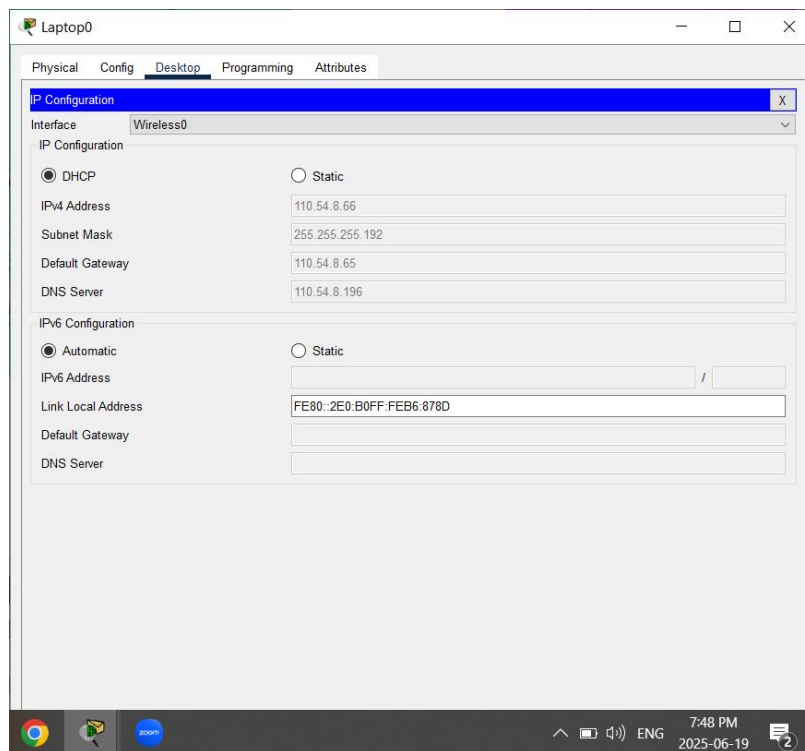


Image 43:Dynamic IP Configuration for Laptop0(In university area)

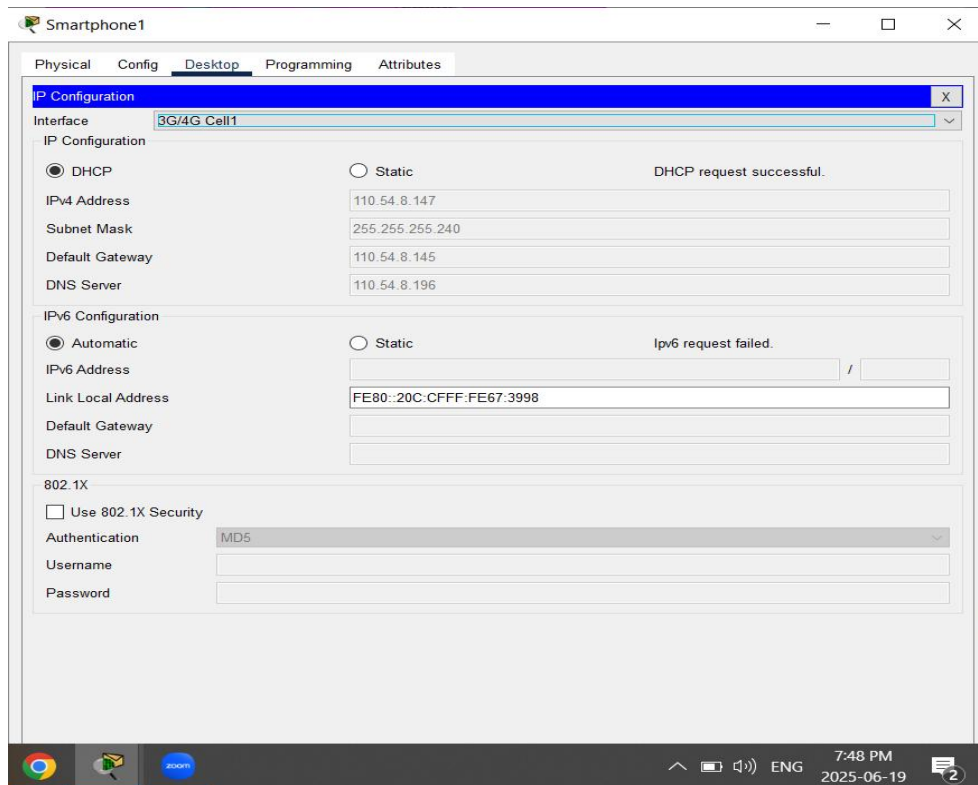


Image 44:Dynamic IP Configuration for Smartphone1(Street area)

Devices in NET1-B and NET2 were configured to obtain IP addresses dynamically via DHCP. Router interfaces were set with ip helper-address where needed to forward DHCP requests to the DHCP server in NET1-A. Dynamic addressing ensures easy scalability and minimal manual configuration.

3. Static IP Configuration for End Devices

Devices such as PCs in the Home network and all servers in the Datacenter were manually configured with static IPs to ensure consistency and accessibility.

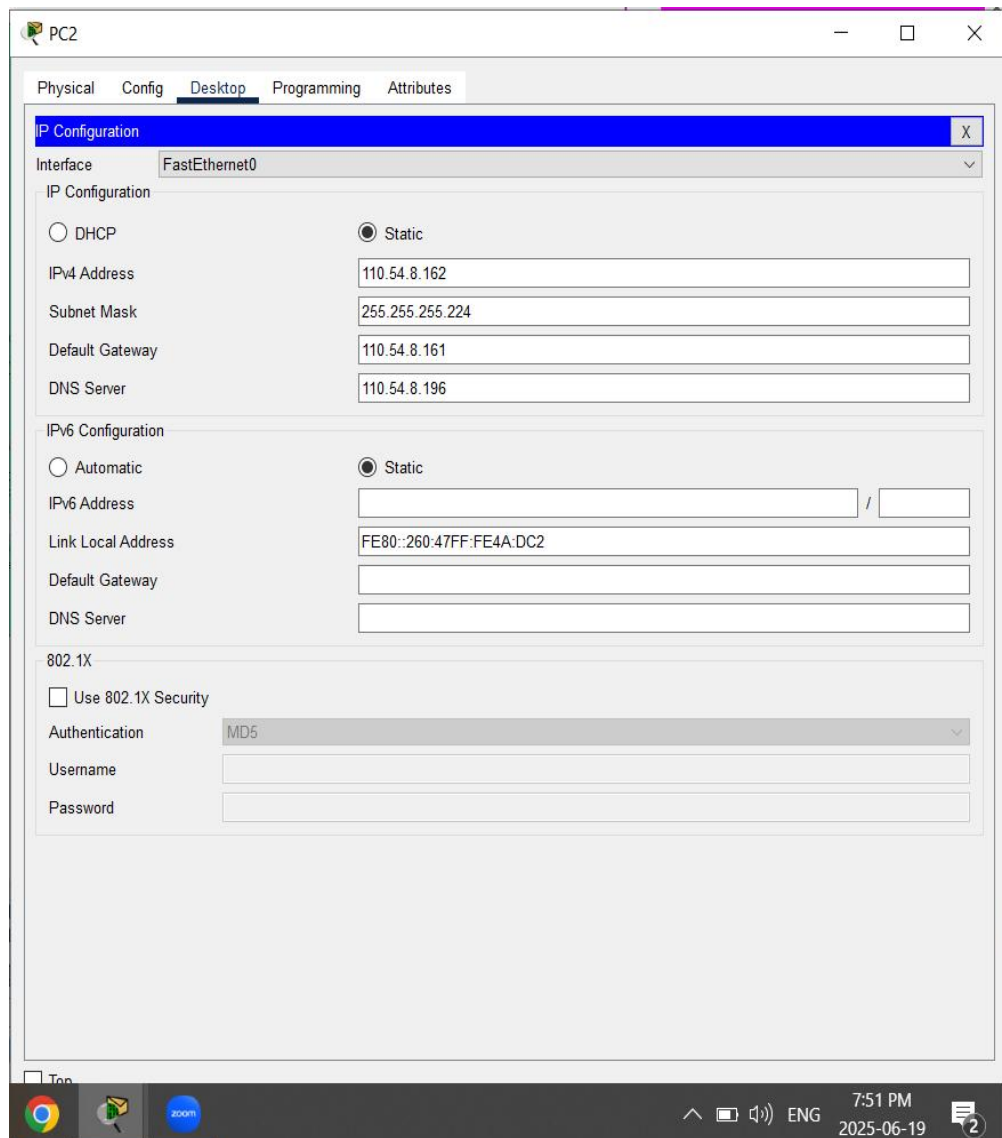


Image 45:3.Static IP Configuration for PC2(in Home Area)

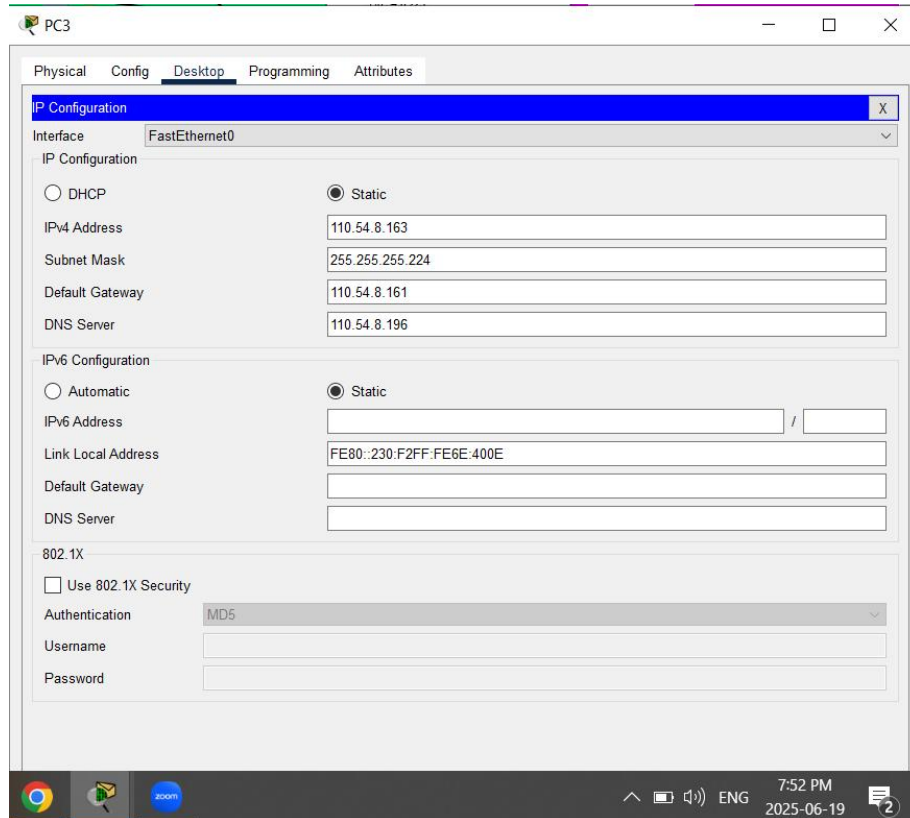


Image 46:Static IP Configuration for PC3(in Home Area)

Home network PCs and Datacenter servers were manually configured with static IP addresses. This approach ensures fixed connectivity and reliability, which is essential for service-hosting devices like the Web, DNS, and Mail servers.

Successful Ping and tracert results between end devices.

Ping and Traceroute commands were used to test end-to-end connectivity between devices across different subnets. These confirm the success of OSPF routing and device accessibility.


```
C:\>ping 110.54.8.147

Pinging 110.54.8.147 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 110.54.8.147: bytes=32 time=46ms TTL=125
Reply from 110.54.8.147: bytes=32 time=58ms TTL=125

Ping statistics for 110.54.8.147:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 46ms, Maximum = 58ms, Average = 52ms

C:\>|
```

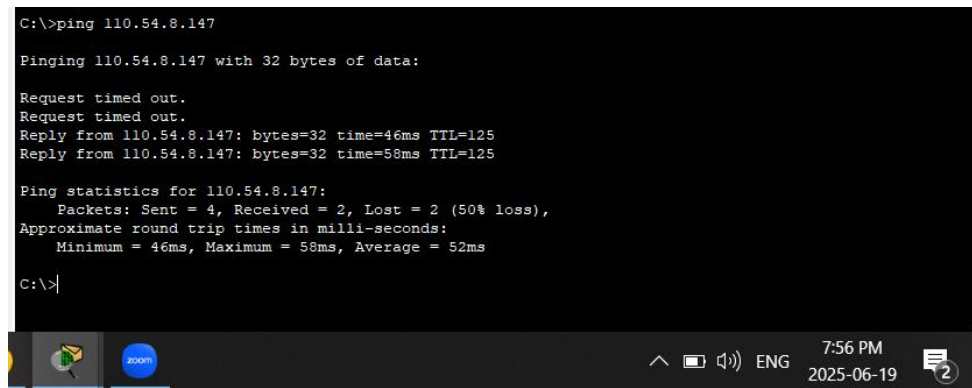


Figure 2:ping from PC at university and a smartphone at the street

```
C:\>tracert 110.54.8.162

Tracing route to 110.54.8.162 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    110.54.8.1
  1  1 ms    1 ms    9 ms    110.54.8.226
  2  *        1 ms   10 ms   110.54.8.162

Trace complete.

C:\>|
```

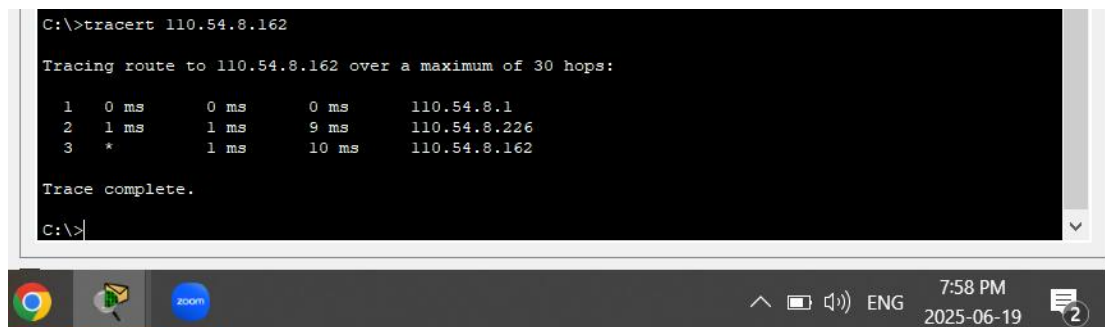


Figure 3: tracert from PC at university and a PC at the Home

```
PC2
Physical Config Desktop Programming Attributes
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 110.54.8.66

Tracing route to 110.54.8.66 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    110.54.8.161
  1  0 ms    0 ms    0 ms    110.54.8.225
  2  *       11 ms   25 ms   110.54.8.66

Trace complete.

C:\>|
```

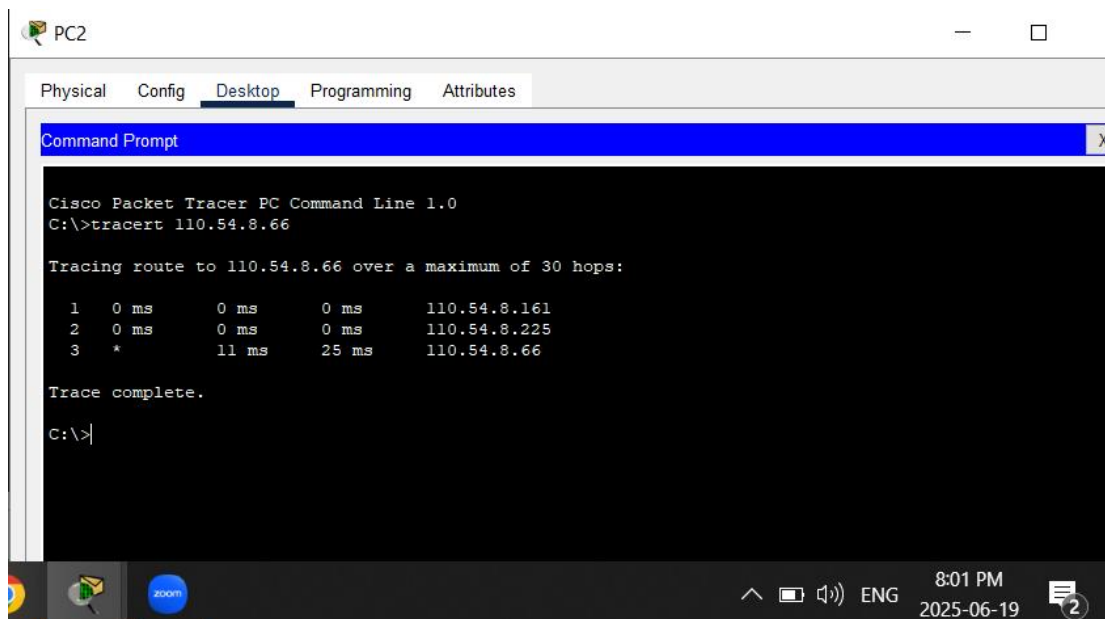


Figure 4: tracert from PC at Home and a laptop at the university

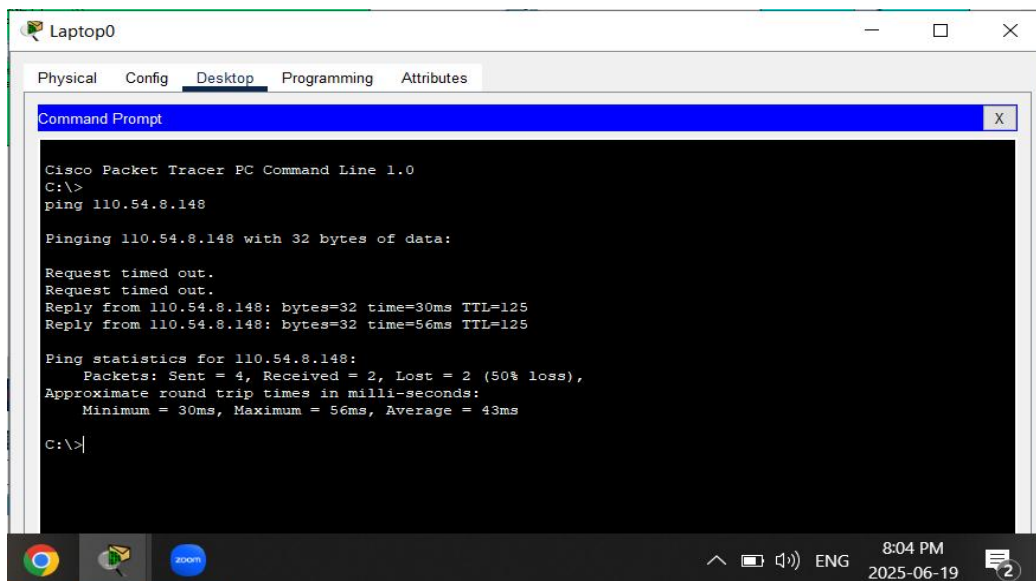


Figure 5:ping a laptop at the university to a smartphone at street

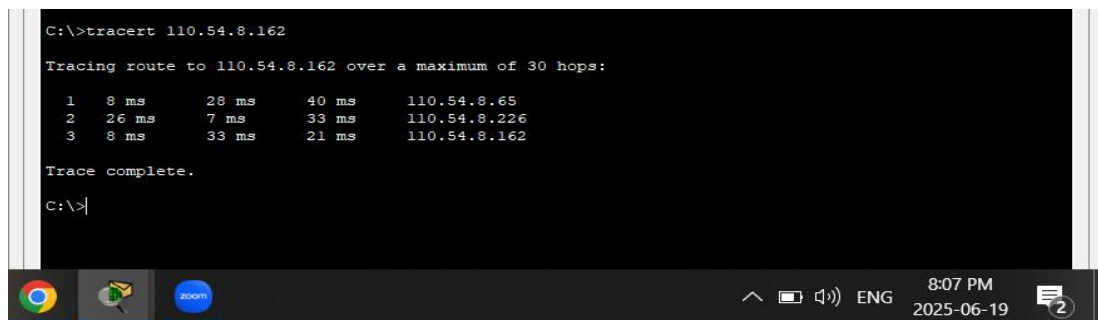


Figure 6:tracert from a laptop at the university to a pc at home

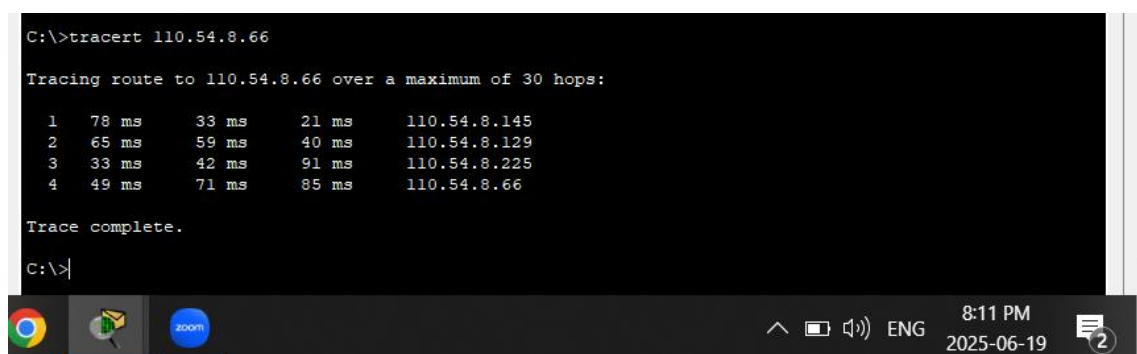


Figure 7: ping from smartphone at street to a laptop at university

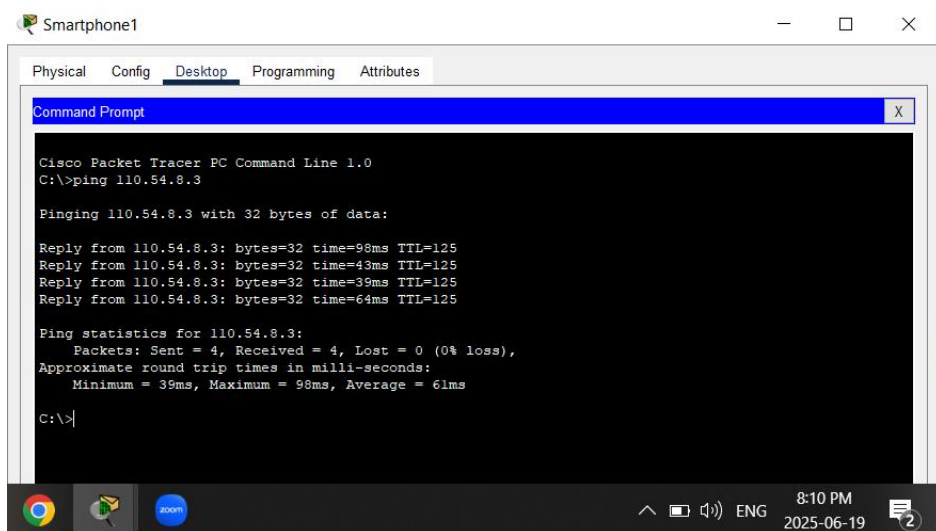


Figure 8: ping from smartphone at street to a pc at university

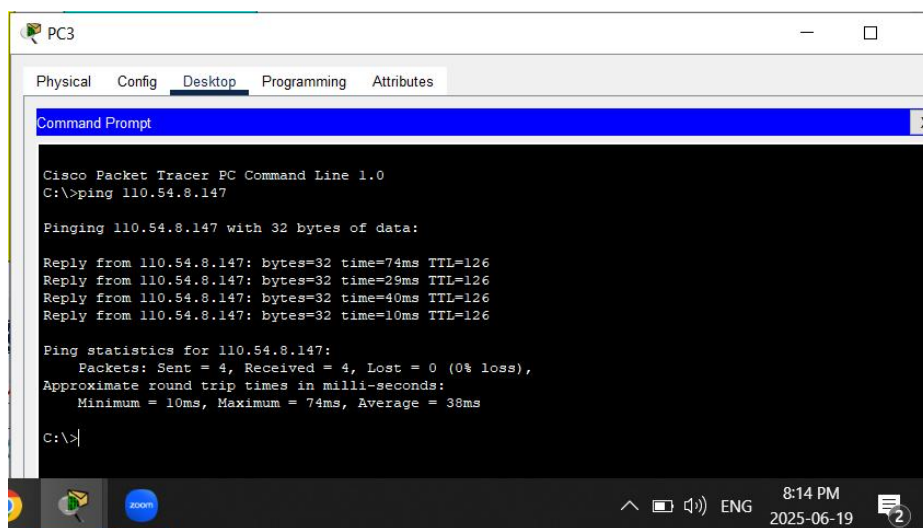


Figure 9: ping from pc at home to a smartphone at street

Ping and traceroute commands were used to verify end-to-end connectivity and routing efficiency. Successful pings confirmed IP reachability, while traceroute showed correct OSPF routing paths through multiple hops. Any failed attempts were corrected by adjusting routing tables or subnet assignments.

Successful Access to the Web Server www.coe.birzeit.edu from End Devices

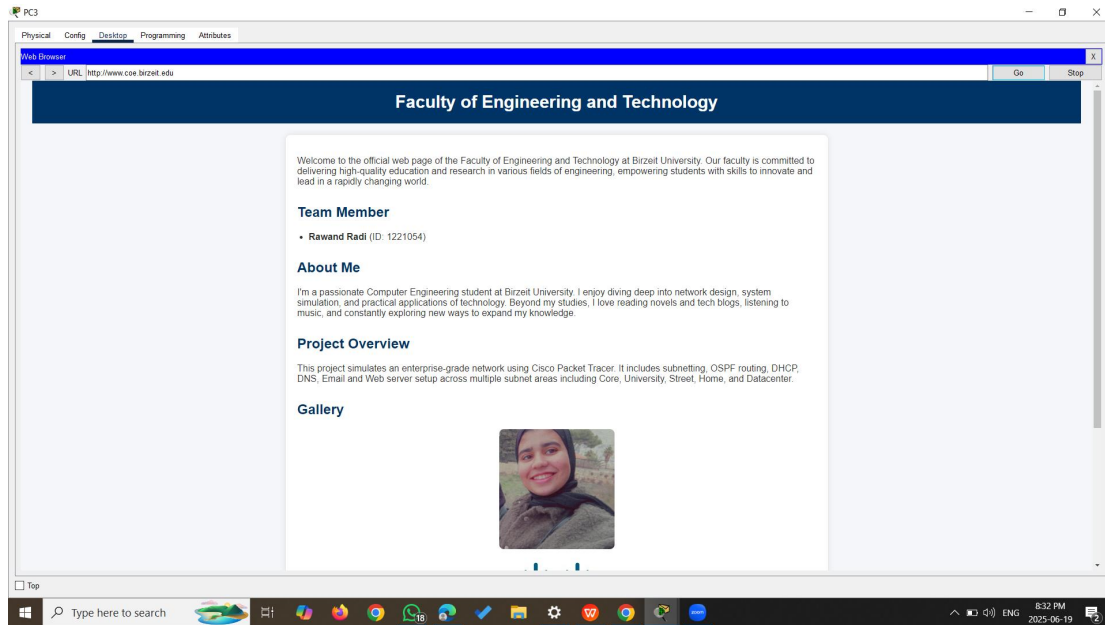


Figure 10: Access webpage from pc at home area

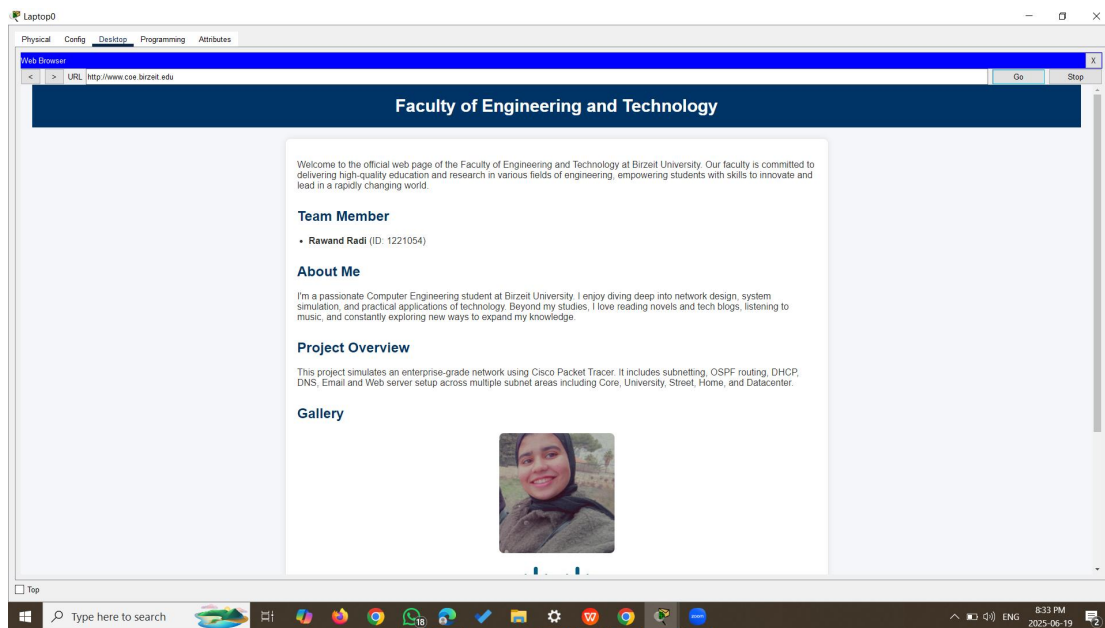


Figure 11: Access webpage from laptop at university area

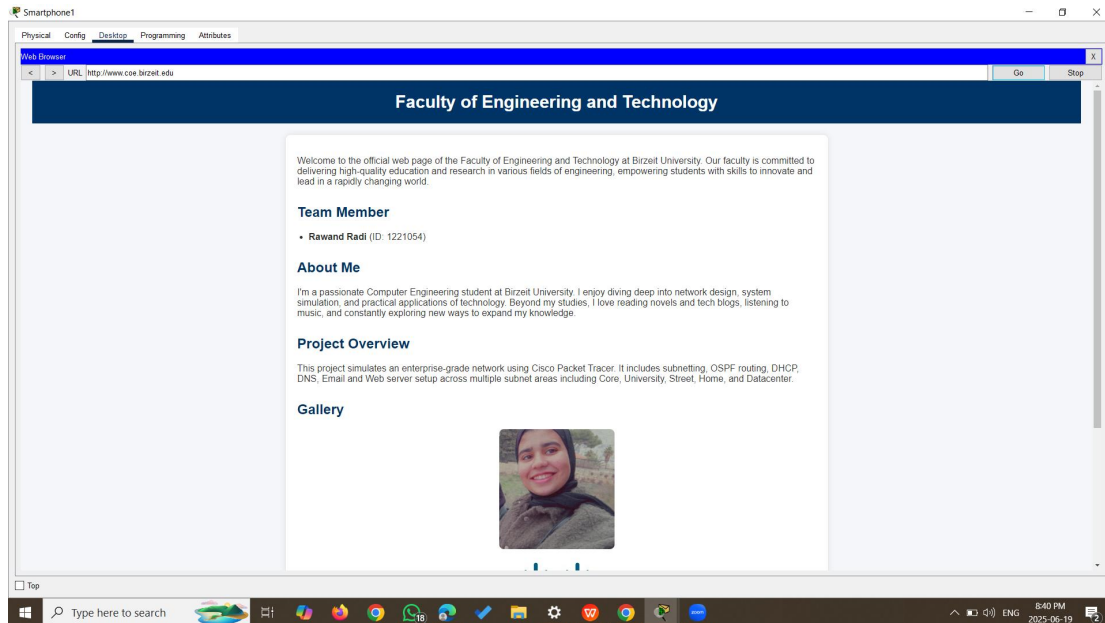


Figure 12: Access webpage from smartphone at street area

PCs from various subnets successfully accessed www.coe.birzeit.edu using a web browser. This verified that the DNS resolution was correct and that the web server was reachable across the OSPF-enabled network.

Successful Sending and Receiving of Emails Between Users

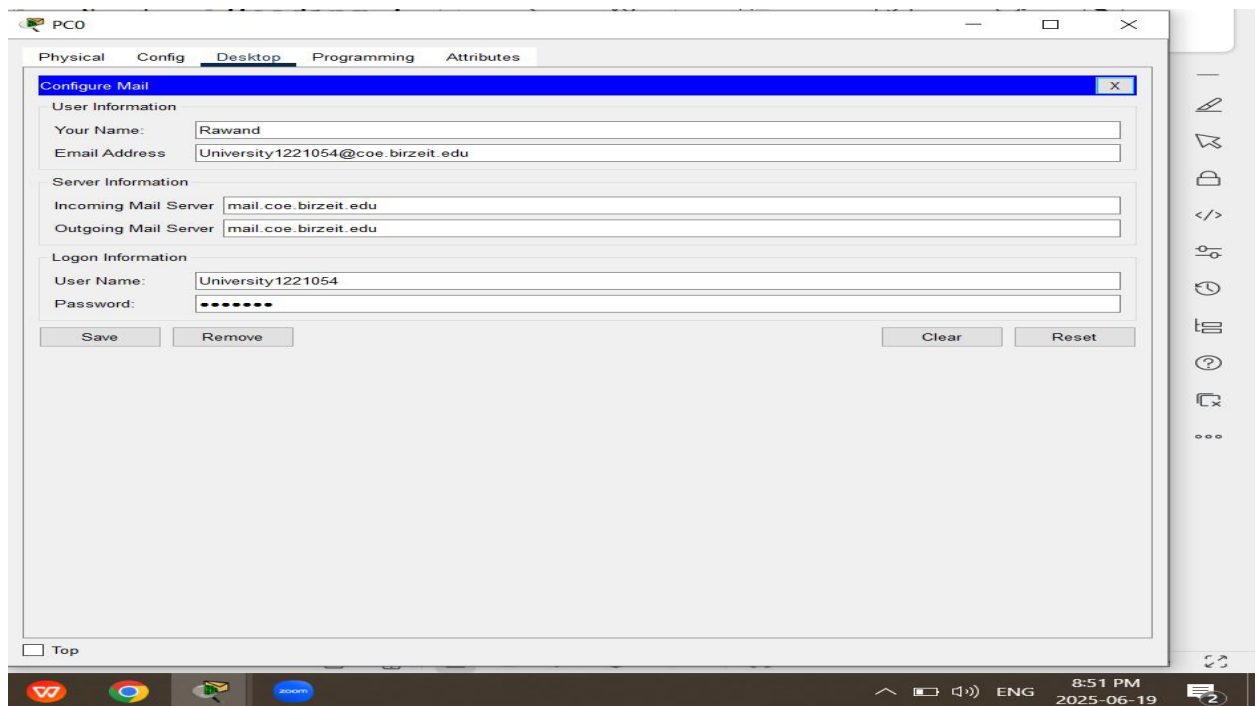


Image 47: Account log in for Mail in PC0 at the university area

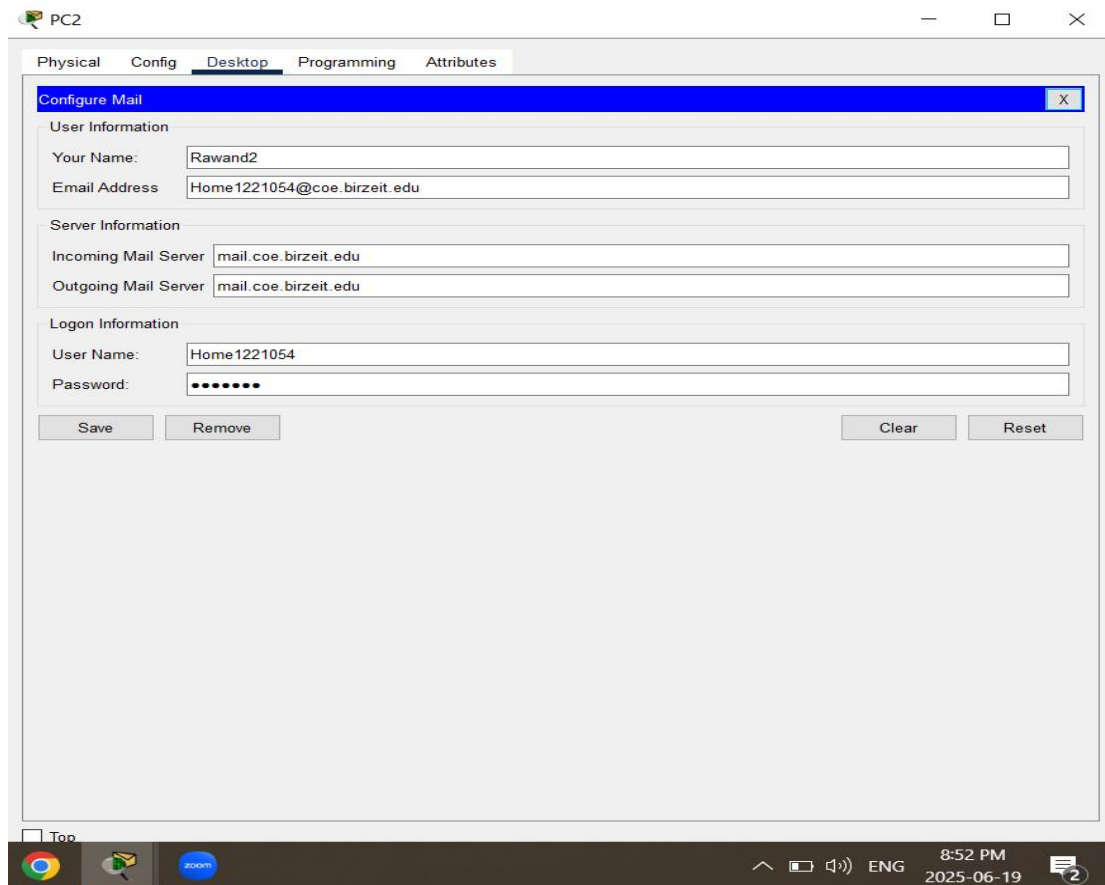


Image 48:Account log in for Mail in PC2 at the Home area

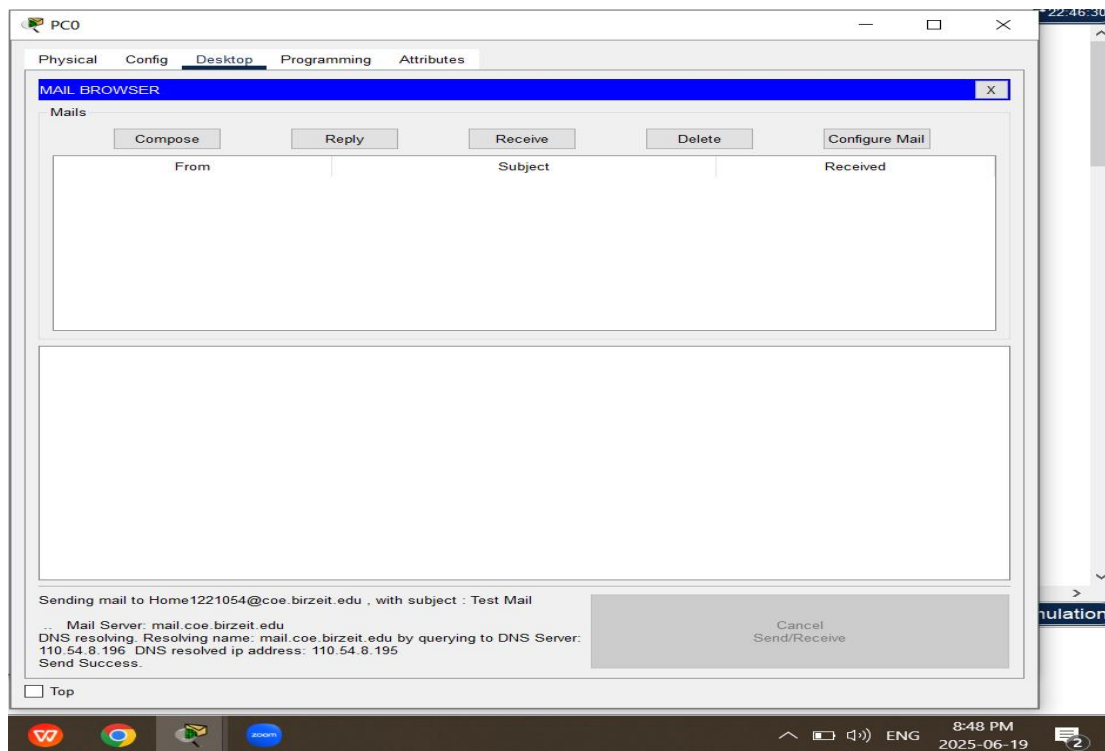


Image 49:Mail send from PC0 at university to OC2 at Home

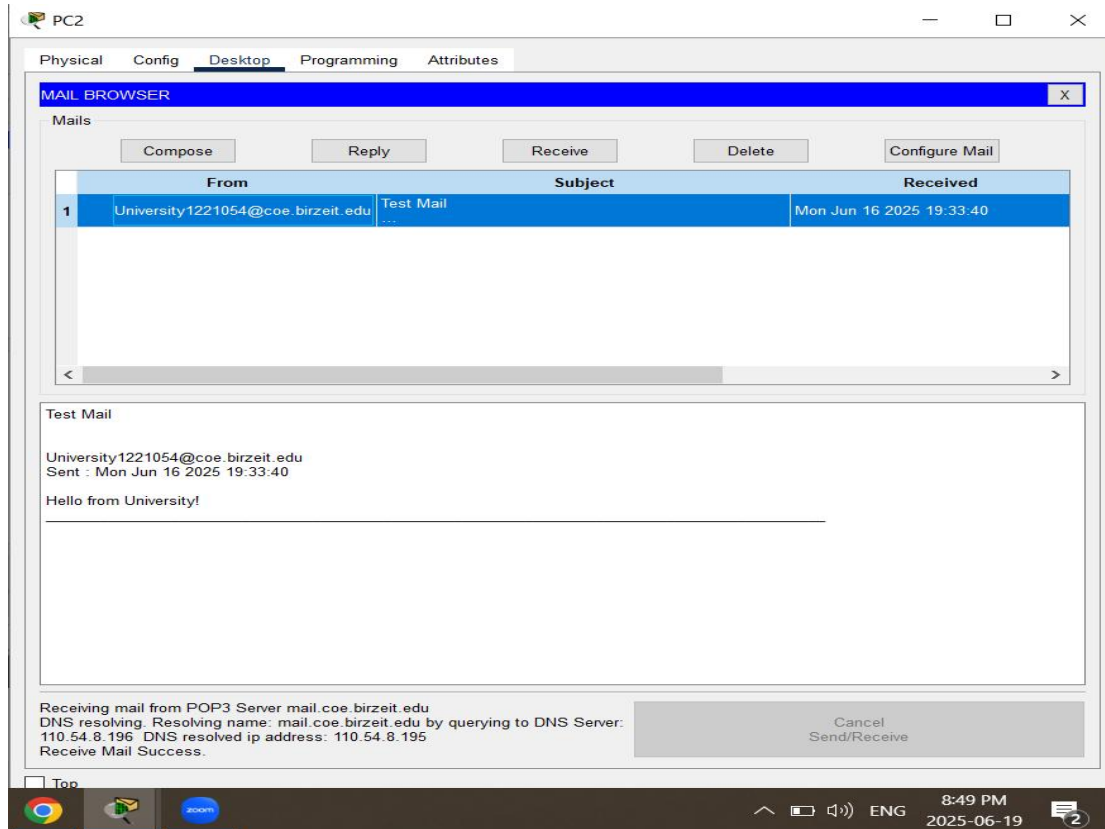


Image 50:Mail Received From PC at PC2(Home Area)

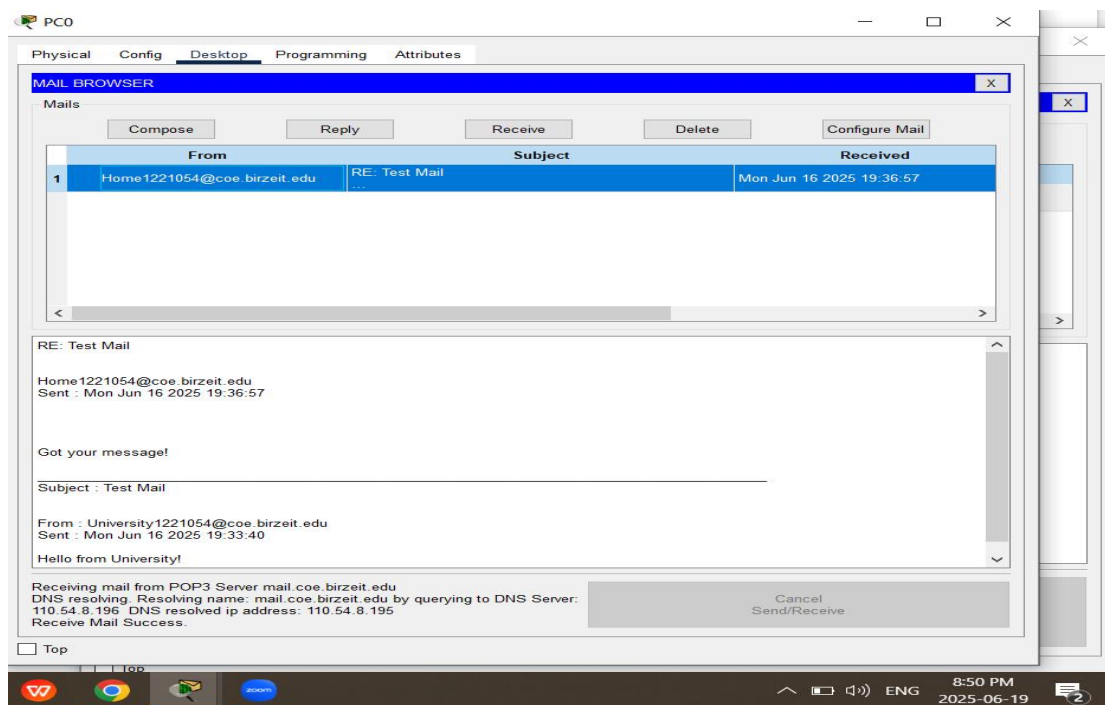


Image 51:Reply mail from PC2 to PC0(University area)

To validate mail server functionality, successful email exchange was tested between end devices located in the Home and University networks. A user from the University PC sent an email to a user in the Home PC via the configured Mail Server

(mail.coe.birzeit.edu). The message was successfully delivered and replied to, confirming that both SMTP and POP3 services were working correctly. Attached screenshots show the full process of sending, receiving, and replying to messages between users.

Issues and Limitations

Several challenges were faced during the project:

1. **DHCP in NET1-B** : Devices in NET1-B couldn't get IPs at first because the DHCP server was in NET1-A. The issue was fixed by adding an ip helper-address on Router0.
2. **Subnetting in NET2** : The Central Office Server had two interfaces, so NET2 had to be split into two /28 subnets instead of one /27. This added complexity to the IP plan.
3. **Static IPs for Some Devices**: Devices like smartphones and home PCs were given static IPs. While this works in Packet Tracer, DHCP would be better in real life for mobile devices.
4. **Email Limitations**: Email testing worked, but Packet Tracer's email system is very basic. It lacks
5. **Time Constraints** : As a solo team member, setting up, testing, and documenting everything took time and effort. Some parts were tested briefly due to limited time.
6. **Tool Limitations**: Packet Tracer has its own limits. Some ping or tracert tests failed even when everything was correctly set up.

Teamwork (Ratios and Details)

This project was completed individually by Rawand Radi (SID: 1221054). As a solo participant, all responsibilities were handled independently, including:

Network Design & Subnetting: 100%

Designed the topology, calculated subnet ranges, and assigned IP addresses.

- **Device Configuration: 100%**
Configured routers, switches, access points, DHCP, DNS, Mail, and Web servers.
- **OSPF Routing Setup: 100%**
Assigned areas and activated OSPF across all routers.
- **Testing and Troubleshooting: 100%**
Conducted ping/tracert tests, server connectivity checks, and resolved issues (like DHCP relay).
- **Documentation & Report Writing: 100%**
Created diagrams, captured screenshots, and wrote all sections of the report, including theory, procedure, results, and discussion.

Despite being a solo project, all requirements were met successfully, and tasks were completed with attention to accuracy and completeness.

Conclusion

This project provided hands-on experience in designing, configuring, and testing a complete enterprise network using Cisco Packet Tracer. Through subnetting, OSPF routing, DHCP, DNS, Web, and Email server configurations, all project requirements were met and demonstrated successfully. Each network area was built with careful IP planning and logical topology design to ensure connectivity and efficiency.

Challenges such as DHCP relay setup and subnet conflicts were identified and resolved, enhancing troubleshooting skills. The project not only strengthened practical networking knowledge but also reinforced the importance of documentation, testing, and network service integration. Overall, the work simulated a real-world networking scenario and was completed successfully with full functionality.

References

- [1] [Introduction to IP addressing and subnetting | TechTarget](#) [Accessed: June 18, 2025 at 14:20]
- [2] [What is Cisco Packet Tracer? | Free Training and Download](#) [Accessed: June 18, 2025 at 15:05]
- [3] [What is DHCP Server in Windows Server? | Microsoft Learn](#) [Accessed: June 18, 2025 at 17:45]
- [4] [What is DNS? | How DNS works | Cloudflare](#) [Accessed: June 19, 2025 at 12:50]
- [5] [What Are Email Protocols - POP3, SMTP and IMAP - SiteGround Tutorials](#) [Accessed: June 19, 2025 at 13:30]
- [6] [Configure Authentication in Open Shortest Path First - Cisco](#) [Accessed: June 18, 2025 at 19:00]